

Model for “kagome”

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General Condition

- Basis type: **lgs**
- SAMB selection:
 - Type: [Q, G]
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_g , A_u , E_g , E_u]
 - Spin (s): [0, 1]
- Atomic selection:
 - Type: [Q, G, M, T]
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_g , A_u , E_g , E_u]
 - Spin (s): [0, 1]
- Site-cluster selection:
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_g , A_u , E_g , E_u]
- Bond-cluster selection:
 - Type: [Q, G, M, T]
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_g , A_u , E_g , E_u]
- Max. neighbor: 10
- Search cell range: (-2, 3), (-2, 3), (-2, 3)
- Toroidal priority: **false**

Group and Unit Cell

- Group: SG No. 147 $C_{3i}^1 P\bar{3}$ [trigonal]
- Associated point group: PG No. 147 $C_{3i} \bar{3}$ [trigonal]
- Unit cell:
 - $a = 1.00000$, $b = 1.00000$, $c = 1.00000$, $\alpha = 90.0$, $\beta = 90.0$, $\gamma = 120.0$
- Lattice vectors (conventional cell):
 - $\mathbf{a}_1 = [1.00000, 0.00000, 0.00000]$
 - $\mathbf{a}_2 = [-0.50000, 0.86603, 0.00000]$
 - $\mathbf{a}_3 = [0.00000, 0.00000, 1.00000]$

Symmetry Operation

Table 1: Symmetry operation

#	SO	#	SO	#	SO	#	SO	#	SO
1	$\{1 0\}$	2	$\{3_{001}^+ 0\}$	3	$\{3_{001}^- 0\}$	4	$\{-1 0\}$	5	$\{-3_{001}^+ 0\}$
6	$\{-3_{001}^- 0\}$								

Harmonics

Table 2: Harmonics

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
1	$\mathbb{Q}_0(A_g)$	A_g	0	Q, T	-	-	1
2	$\mathbb{G}_1(A_g)$	A_g	1	G, M	-	-	z
3	$\mathbb{Q}_2(A_g)$	A_g	2	Q, T	-	-	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$
4	$\mathbb{G}_3(A_g, 1)$	A_g	3	G, M	1	-	$-\frac{z(3x^2+3y^2-2z^2)}{2}$
5	$\mathbb{G}_3(A_g, 2)$	A_g	3	G, M	2	-	$\frac{\sqrt{10}y(3x^2-y^2)}{4}$
6	$\mathbb{G}_3(A_g, 3)$	A_g	3	G, M	3	-	$\frac{\sqrt{10}x(x^2-3y^2)}{4}$
7	$\mathbb{Q}_4(A_g, 2)$	A_g	4	Q, T	2	-	$\frac{\sqrt{70}xz(x^2-3y^2)}{4}$
8	$\mathbb{Q}_4(A_g, 3)$	A_g	4	Q, T	3	-	$\frac{\sqrt{70}yz(3x^2-y^2)}{4}$
9	$\mathbb{G}_0(A_u)$	A_u	0	G, M	-	-	1

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
10	$\mathbb{Q}_1(A_u)$	A_u	1	Q, T	-	-	z
11	$\mathbb{G}_2(A_u)$	A_u	2	G, M	-	-	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$
12	$\mathbb{Q}_3(A_u, 1)$	A_u	3	Q, T	1	-	$-\frac{z(3x^2+3y^2-2z^2)}{2}$
13	$\mathbb{Q}_3(A_u, 2)$	A_u	3	Q, T	2	-	$\frac{\sqrt{10}y(3x^2-y^2)}{4}$
14	$\mathbb{Q}_3(A_u, 3)$	A_u	3	Q, T	3	-	$\frac{\sqrt{10}x(x^2-3y^2)}{4}$
15	$\mathbb{G}_4(A_u, 1)$	A_u	4	G, M	1	-	$\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$
16	$\mathbb{G}_4(A_u, 2)$	A_u	4	G, M	2	-	$\frac{\sqrt{70}xz(x^2-3y^2)}{4}$
17	$\mathbb{G}_4(A_u, 3)$	A_u	4	G, M	3	-	$\frac{\sqrt{70}yz(3x^2-y^2)}{4}$
18	$\mathbb{G}_{1,1}(E_g)$	E_g	1	G, M	-	1	x
19	$\mathbb{G}_{1,2}(E_g)$					2	y
20	$\mathbb{Q}_{2,1}(E_g, 1)$	E_g	2	Q, T	1	1	$\sqrt{3}yz$
21	$\mathbb{Q}_{2,2}(E_g, 1)$					2	$-\sqrt{3}xz$
22	$\mathbb{Q}_{2,1}(E_g, 2)$	E_g	2	Q, T	2	1	$\frac{\sqrt{3}(x^2-y^2)}{2}$
23	$\mathbb{Q}_{2,2}(E_g, 2)$					2	$-\sqrt{3}xy$
24	$\mathbb{G}_{3,1}(E_g, 1)$	E_g	3	G, M	1	1	$\frac{\sqrt{6}x(-x^2-y^2+4z^2)}{4}$
25	$\mathbb{G}_{3,2}(E_g, 1)$					2	$\frac{\sqrt{6}y(-x^2-y^2+4z^2)}{4}$
26	$\mathbb{G}_{3,1}(E_g, 2)$	E_g	3	G, M	2	1	$\sqrt{15}xyz$
27	$\mathbb{G}_{3,2}(E_g, 2)$					2	$\frac{\sqrt{15}z(x^2-y^2)}{2}$
28	$\mathbb{Q}_{4,1}(E_g, 1)$	E_g	4	Q, T	1	1	$\frac{\sqrt{10}yz(-3x^2-3y^2+4z^2)}{4}$
29	$\mathbb{Q}_{4,2}(E_g, 1)$					2	$\frac{\sqrt{10}xz(3x^2+3y^2-4z^2)}{4}$
30	$\mathbb{Q}_{4,1}(E_g, 2)$	E_g	4	Q, T	2	1	$\frac{\sqrt{35}(x^4-6x^2y^2+y^4)}{8}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
31	$\mathbb{Q}_{4,2}(E_g, 2)$					2	$\frac{\sqrt{35}xy(x^2-y^2)}{2}$
32	$\mathbb{Q}_{4,1}(E_g, 3)$	E_g	4	Q, T	3	1	$\frac{\sqrt{5}(-x^4+6x^2z^2+y^4-6y^2z^2)}{4}$
33	$\mathbb{Q}_{4,2}(E_g, 3)$					2	$\frac{\sqrt{5}xy(x^2+y^2-6z^2)}{2}$
34	$\mathbb{Q}_{1,1}(E_u)$	E_u	1	Q, T	-	1	x
35	$\mathbb{Q}_{1,2}(E_u)$					2	y
36	$\mathbb{G}_{2,1}(E_u, 1)$	E_u	2	G, M	1	1	$\sqrt{3}yz$
37	$\mathbb{G}_{2,2}(E_u, 1)$					2	$-\sqrt{3}xz$
38	$\mathbb{G}_{2,1}(E_u, 2)$	E_u	2	G, M	2	1	$\frac{\sqrt{3}(x^2-y^2)}{2}$
39	$\mathbb{G}_{2,2}(E_u, 2)$					2	$-\sqrt{3}xy$
40	$\mathbb{Q}_{3,1}(E_u, 1)$	E_u	3	Q, T	1	1	$\frac{\sqrt{6}x(-x^2-y^2+4z^2)}{4}$
41	$\mathbb{Q}_{3,2}(E_u, 1)$					2	$\frac{\sqrt{6}y(-x^2-y^2+4z^2)}{4}$
42	$\mathbb{Q}_{3,1}(E_u, 2)$	E_u	3	Q, T	2	1	$\sqrt{15}xyz$
43	$\mathbb{Q}_{3,2}(E_u, 2)$					2	$\frac{\sqrt{15}z(x^2-y^2)}{2}$
44	$\mathbb{G}_{4,1}(E_u, 1)$	E_u	4	G, M	1	1	$\frac{\sqrt{10}yz(-3x^2-3y^2+4z^2)}{4}$
45	$\mathbb{G}_{4,2}(E_u, 1)$					2	$\frac{\sqrt{10}xz(3x^2+3y^2-4z^2)}{4}$
46	$\mathbb{G}_{4,1}(E_u, 2)$	E_u	4	G, M	2	1	$\frac{\sqrt{35}(x^4-6x^2y^2+y^4)}{8}$
47	$\mathbb{G}_{4,2}(E_u, 2)$					2	$\frac{\sqrt{35}xy(x^2-y^2)}{2}$
48	$\mathbb{G}_{4,1}(E_u, 3)$	E_u	4	G, M	3	1	$\frac{\sqrt{5}(-x^4+6x^2z^2+y^4-6y^2z^2)}{4}$
49	$\mathbb{G}_{4,2}(E_u, 3)$					2	$\frac{\sqrt{5}xy(x^2+y^2-6z^2)}{2}$

 Basis in full matrix

Table 3: dimension = 24

#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
0	$ s, \uparrow\rangle @A(1)$	1	$ s, \downarrow\rangle @A(1)$	2	$ p_x, \uparrow\rangle @A(1)$	3	$ p_x, \downarrow\rangle @A(1)$	4	$ p_y, \uparrow\rangle @A(1)$
5	$ p_y, \downarrow\rangle @A(1)$	6	$ p_z, \uparrow\rangle @A(1)$	7	$ p_z, \downarrow\rangle @A(1)$	8	$ s, \uparrow\rangle @A(2)$	9	$ s, \downarrow\rangle @A(2)$
10	$ p_x, \uparrow\rangle @A(2)$	11	$ p_x, \downarrow\rangle @A(2)$	12	$ p_y, \uparrow\rangle @A(2)$	13	$ p_y, \downarrow\rangle @A(2)$	14	$ p_z, \uparrow\rangle @A(2)$
15	$ p_z, \downarrow\rangle @A(2)$	16	$ s, \uparrow\rangle @A(3)$	17	$ s, \downarrow\rangle @A(3)$	18	$ p_x, \uparrow\rangle @A(3)$	19	$ p_x, \downarrow\rangle @A(3)$
20	$ p_y, \uparrow\rangle @A(3)$	21	$ p_y, \downarrow\rangle @A(3)$	22	$ p_z, \uparrow\rangle @A(3)$	23	$ p_z, \downarrow\rangle @A(3)$		

Table 4: Atomic basis (orbital part only)

orbital	definition
$ s\rangle$	1
$ p_x\rangle$	x
$ p_y\rangle$	y
$ p_z\rangle$	z

— SAMB: 468 (all 468)

- **A** : 'A' site-cluster

- * bra: $\langle s, \uparrow|$, $\langle s, \downarrow|$

- * ket: $|s, \uparrow\rangle$, $|s, \downarrow\rangle$

- * wyckoff: **3e**

z1

 $\mathbb{Q}_0^{(c)}(A_g) = \mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_0^{(s)}(A_g)$

$$\boxed{\text{z157}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,1}^{(s)}(E_g, 1)}{2}$$

$$\boxed{\text{z158}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,2}^{(s)}(E_g, 1)}{2}$$

• **A** : 'A' site-cluster

* bra: $\langle s, \uparrow |, \langle s, \downarrow |$

* ket: $|p_x, \uparrow\rangle, |p_x, \downarrow\rangle, |p_y, \uparrow\rangle, |p_y, \downarrow\rangle, |p_z, \uparrow\rangle, |p_z, \downarrow\rangle$

* wyckoff: **3e**

$$\boxed{\text{z81}} \quad \mathbb{Q}_1^{(c)}(A_u, a) = \mathbb{Q}_1^{(a)}(A_u)\mathbb{Q}_0^{(s)}(A_g)$$

$$\boxed{\text{z82}} \quad \mathbb{Q}_1^{(c)}(A_u, b) = \frac{\sqrt{2} \left(-\mathbb{Q}_{1,1}^{(a)}(E_u)\mathbb{Q}_{2,2}^{(s)}(E_g, 1) + \mathbb{Q}_{1,2}^{(a)}(E_u)\mathbb{Q}_{2,1}^{(s)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z83}} \quad \mathbb{Q}_1^{(1,-1;c)}(A_u) = \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{2,2}^{(s)}(E_g, 1) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{2,1}^{(s)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z84}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, 2) = \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{2,2}^{(s)}(E_g, 1) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{2,1}^{(s)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z85}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, 3) = \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{2,1}^{(s)}(E_g, 1) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{2,2}^{(s)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z86}} \quad \mathbb{Q}_1^{(1,0;c)}(A_u, a) = \mathbb{Q}_1^{(1,0;a)}(A_u)\mathbb{Q}_0^{(s)}(A_g)$$

$$\boxed{\text{z87}} \quad \mathbb{Q}_1^{(1,0;c)}(A_u, b) = \frac{\sqrt{2} \left(-\mathbb{Q}_{1,1}^{(1,0;a)}(E_u)\mathbb{Q}_{2,2}^{(s)}(E_g, 1) + \mathbb{Q}_{1,2}^{(1,0;a)}(E_u)\mathbb{Q}_{2,1}^{(s)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z88}} \quad \mathbb{G}_2^{(c)}(A_u) = \frac{\sqrt{2} \left(\mathbb{Q}_{1,1}^{(a)}(E_u)\mathbb{Q}_{2,1}^{(s)}(E_g, 1) + \mathbb{Q}_{1,2}^{(a)}(E_u)\mathbb{Q}_{2,2}^{(s)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z89}} \quad \mathbb{G}_0^{(1,-1;c)}(A_u) = \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{2,1}^{(s)}(E_g, 1) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{2,2}^{(s)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z90}} \quad \mathbb{G}_2^{(1,-1;c)}(A_u) = \mathbb{G}_2^{(1,-1;a)}(A_u)\mathbb{Q}_0^{(s)}(A_g)$$

$$\boxed{\text{z91}} \quad \mathbb{G}_2^{(1,0;c)}(A_u) = \frac{\sqrt{2} \left(\mathbb{Q}_{1,1}^{(1,0;a)}(E_u) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) + \mathbb{Q}_{1,2}^{(1,0;a)}(E_u) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z92}} \quad \mathbb{G}_0^{(1,1;c)}(A_u) = \mathbb{G}_0^{(1,1;a)}(A_u) \mathbb{Q}_0^{(s)}(A_g)$$

$$\boxed{\text{z317}} \quad \mathbb{Q}_{1,1}^{(c)}(E_u, a) = \frac{\sqrt{2} \mathbb{Q}_{1,1}^{(a)}(E_u) \mathbb{Q}_0^{(s)}(A_g)}{2}$$

$$\boxed{\text{z318}} \quad \mathbb{Q}_{1,2}^{(c)}(E_u, a) = \frac{\sqrt{2} \mathbb{Q}_{1,2}^{(a)}(E_u) \mathbb{Q}_0^{(s)}(A_g)}{2}$$

$$\boxed{\text{z319}} \quad \mathbb{Q}_{1,1}^{(c)}(E_u, b) = -\frac{\sqrt{2} \mathbb{Q}_1^{(a)}(A_u) \mathbb{Q}_{2,2}^{(s)}(E_g, 1)}{2}$$

$$\boxed{\text{z320}} \quad \mathbb{Q}_{1,2}^{(c)}(E_u, b) = \frac{\sqrt{2} \mathbb{Q}_1^{(a)}(A_u) \mathbb{Q}_{2,1}^{(s)}(E_g, 1)}{2}$$

$$\boxed{\text{z321}} \quad \mathbb{Q}_{3,1}^{(c)}(E_u, 2) = \frac{\mathbb{Q}_{1,1}^{(a)}(E_u) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) - \mathbb{Q}_{1,2}^{(a)}(E_u) \mathbb{Q}_{2,2}^{(s)}(E_g, 1)}{2}$$

$$\boxed{\text{z322}} \quad \mathbb{Q}_{3,2}^{(c)}(E_u, 2) = -\frac{\mathbb{Q}_{1,1}^{(a)}(E_u) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) + \mathbb{Q}_{1,2}^{(a)}(E_u) \mathbb{Q}_{2,1}^{(s)}(E_g, 1)}{2}$$

$$\boxed{\text{z323}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(E_u) = \frac{\sqrt{10} \left(-\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) - \sqrt{3} \mathbb{G}_2^{(1,-1;a)}(A_u) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) \right)}{10}$$

$$\boxed{\text{z324}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{10} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) - \sqrt{3} \mathbb{G}_2^{(1,-1;a)}(A_u) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) \right)}{10}$$

$$\boxed{\text{z325}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 1) = \frac{\sqrt{5} \left(\sqrt{3} \mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) - \sqrt{3} \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) - 2 \mathbb{G}_2^{(1,-1;a)}(A_u) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) \right)}{10}$$

$$\boxed{\text{z326}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 1) = -\frac{\sqrt{5} \left(\sqrt{3} \mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) + \sqrt{3} \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) + 2 \mathbb{G}_2^{(1,-1;a)}(A_u) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) \right)}{10}$$

$$\boxed{\text{z327}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(E_u, a) = \frac{\sqrt{2} \mathbb{Q}_{1,1}^{(1,0;a)}(E_u) \mathbb{Q}_0^{(s)}(A_g)}{2}$$

$$\boxed{\text{z328}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(E_u, a) = \frac{\sqrt{2} \mathbb{Q}_{1,2}^{(1,0;a)}(E_u) \mathbb{Q}_0^{(s)}(A_g)}{2}$$

$$\boxed{\text{z329}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(E_u, b) = -\frac{\sqrt{2}\mathbb{Q}_1^{(1,0;a)}(A_u)\mathbb{Q}_{2,2}^{(s)}(E_g, 1)}{2}$$

$$\boxed{\text{z330}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(E_u, b) = \frac{\sqrt{2}\mathbb{Q}_1^{(1,0;a)}(A_u)\mathbb{Q}_{2,1}^{(s)}(E_g, 1)}{2}$$

$$\boxed{\text{z331}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 2) = \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E_u)\mathbb{Q}_{2,1}^{(s)}(E_g, 1) - \mathbb{Q}_{1,2}^{(1,0;a)}(E_u)\mathbb{Q}_{2,2}^{(s)}(E_g, 1)}{2}$$

$$\boxed{\text{z332}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 2) = -\frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E_u)\mathbb{Q}_{2,2}^{(s)}(E_g, 1) + \mathbb{Q}_{1,2}^{(1,0;a)}(E_u)\mathbb{Q}_{2,1}^{(s)}(E_g, 1)}{2}$$

$$\boxed{\text{z333}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_g)}{2}$$

$$\boxed{\text{z334}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(s)}(A_g)}{2}$$

$$\boxed{\text{z335}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, 2a) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_g)}{2}$$

$$\boxed{\text{z336}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, 2a) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_0^{(s)}(A_g)}{2}$$

$$\boxed{\text{z337}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, 2b) = \frac{-\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{2,1}^{(s)}(E_g, 1) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{2,2}^{(s)}(E_g, 1)}{2}$$

$$\boxed{\text{z338}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, 2b) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{2,2}^{(s)}(E_g, 1) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{2,1}^{(s)}(E_g, 1)}{2}$$

$$\boxed{\text{z339}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{2,1}^{(s)}(E_g, 1)}{2}$$

$$\boxed{\text{z340}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{2,2}^{(s)}(E_g, 1)}{2}$$

• **A : 'A' site-cluster**

* bra: $\langle p_x, \uparrow |, \langle p_x, \downarrow |, \langle p_y, \uparrow |, \langle p_y, \downarrow |, \langle p_z, \uparrow |, \langle p_z, \downarrow |$

* ket: $|p_x, \uparrow \rangle, |p_x, \downarrow \rangle, |p_y, \uparrow \rangle, |p_y, \downarrow \rangle, |p_z, \uparrow \rangle, |p_z, \downarrow \rangle$

* wyckoff: 3e

$$\boxed{\text{z2}} \quad \mathbb{Q}_0^{(c)}(A_g, a) = \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_0^{(s)}(A_g)$$

$$\boxed{\text{z3}} \quad \mathbb{Q}_0^{(c)}(A_g, b) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g, 1) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) + \mathbb{Q}_{2,2}^{(a)}(E_g, 1) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z4}} \quad \mathbb{Q}_2^{(c)}(A_g) = \mathbb{Q}_2^{(a)}(A_g) \mathbb{Q}_0^{(s)}(A_g)$$

$$\boxed{\text{z5}} \quad \mathbb{Q}_4^{(c)}(A_g, 2) = \frac{\sqrt{2} \left(-\mathbb{Q}_{2,1}^{(a)}(E_g, 2) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) + \mathbb{Q}_{2,2}^{(a)}(E_g, 2) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z6}} \quad \mathbb{Q}_4^{(c)}(A_g, 3) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g, 2) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) + \mathbb{Q}_{2,2}^{(a)}(E_g, 2) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z7}} \quad \mathbb{Q}_0^{(1,-1;c)}(A_g) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z8}} \quad \mathbb{Q}_2^{(1,-1;c)}(A_g) = \mathbb{Q}_2^{(1,-1;a)}(A_g) \mathbb{Q}_0^{(s)}(A_g)$$

$$\boxed{\text{z9}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_g, 2) = \frac{\sqrt{2} \left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z10}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_g, 3) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z11}} \quad \mathbb{Q}_2^{(1,0;c)}(A_g) = \frac{\sqrt{2} \left(\mathbb{G}_{1,1}^{(1,0;a)}(E_g) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) + \mathbb{G}_{1,2}^{(1,0;a)}(E_g) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z12}} \quad \mathbb{Q}_0^{(1,1;c)}(A_g) = \mathbb{Q}_0^{(1,1;a)}(A_g) \mathbb{Q}_0^{(s)}(A_g)$$

$$\boxed{\text{z13}} \quad \mathbb{G}_1^{(c)}(A_g) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g, 1) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) - \mathbb{Q}_{2,2}^{(a)}(E_g, 1) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z14}} \quad \mathbb{G}_1^{(1,-1;c)}(A_g) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z15}} \quad \mathbb{G}_1^{(1,0;c)}(A_g, a) = \mathbb{G}_1^{(1,0;a)}(A_g) \mathbb{Q}_0^{(s)}(A_g)$$

$$\begin{aligned}
\text{z16} \quad \mathbb{G}_1^{(1,0;c)}(A_g, b) &= \frac{\sqrt{2} \left(-\mathbb{G}_{1,1}^{(1,0;a)}(E_g) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) + \mathbb{G}_{1,2}^{(1,0;a)}(E_g) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) \right)}{2} \\
\text{z159} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, 1a) &= \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{2,1}^{(s)}(E_g, 1)}{2} \\
\text{z160} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, 1a) &= \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{2,2}^{(s)}(E_g, 1)}{2} \\
\text{z161} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, 1b) &= \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(a)}(E_g, 1) \mathbb{Q}_0^{(s)}(A_g)}{2} \\
\text{z162} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, 1b) &= \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(a)}(E_g, 1) \mathbb{Q}_0^{(s)}(A_g)}{2} \\
\text{z163} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, 1c) &= \frac{\sqrt{14} \left(-\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(E_g, 2) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) + \sqrt{3} \mathbb{Q}_{2,2}^{(a)}(E_g, 2) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) + \mathbb{Q}_2^{(a)}(A_g) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) \right)}{14} \\
\text{z164} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, 1c) &= \frac{\sqrt{14} \left(\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(E_g, 2) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) + \sqrt{3} \mathbb{Q}_{2,2}^{(a)}(E_g, 2) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) + \mathbb{Q}_2^{(a)}(A_g) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) \right)}{14} \\
\text{z165} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, 2a) &= \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(a)}(E_g, 2) \mathbb{Q}_0^{(s)}(A_g)}{2} \\
\text{z166} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, 2a) &= \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(a)}(E_g, 2) \mathbb{Q}_0^{(s)}(A_g)}{2} \\
\text{z167} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, 2b) &= \frac{-\mathbb{Q}_{2,1}^{(a)}(E_g, 1) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) + \mathbb{Q}_{2,2}^{(a)}(E_g, 1) \mathbb{Q}_{2,2}^{(s)}(E_g, 1)}{2} \\
\text{z168} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, 2b) &= \frac{\mathbb{Q}_{2,1}^{(a)}(E_g, 1) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) + \mathbb{Q}_{2,2}^{(a)}(E_g, 1) \mathbb{Q}_{2,1}^{(s)}(E_g, 1)}{2} \\
\text{z169} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 1) &= \frac{\sqrt{7} \left(\mathbb{Q}_{2,1}^{(a)}(E_g, 2) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) - \mathbb{Q}_{2,2}^{(a)}(E_g, 2) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) + 2\sqrt{3} \mathbb{Q}_2^{(a)}(A_g) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) \right)}{14} \\
\text{z170} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 1) &= \frac{\sqrt{7} \left(-\mathbb{Q}_{2,1}^{(a)}(E_g, 2) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) - \mathbb{Q}_{2,2}^{(a)}(E_g, 2) \mathbb{Q}_{2,1}^{(s)}(E_g, 1) + 2\sqrt{3} \mathbb{Q}_2^{(a)}(A_g) \mathbb{Q}_{2,2}^{(s)}(E_g, 1) \right)}{14}
\end{aligned}$$

$$\begin{aligned}
\text{z171} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_g)}{2} \\
\text{z172} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_0^{(s)}(A_g)}{2} \\
\text{z173} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, 1b) &= \frac{\sqrt{14}\left(-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{2,1}^{(s)}(E_g, 1) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{2,2}^{(s)}(E_g, 1) + \mathbb{Q}_2^{(1,-1;a)}(A_g)\mathbb{Q}_{2,1}^{(s)}(E_g, 1)\right)}{14} \\
\text{z174} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, 1b) &= \frac{\sqrt{14}\left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{2,2}^{(s)}(E_g, 1) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{2,1}^{(s)}(E_g, 1) + \mathbb{Q}_2^{(1,-1;a)}(A_g)\mathbb{Q}_{2,2}^{(s)}(E_g, 1)\right)}{14} \\
\text{z175} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, 2a) &= \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_g)}{2} \\
\text{z176} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, 2a) &= \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_0^{(s)}(A_g)}{2} \\
\text{z177} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, 2b) &= \frac{-\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_{2,1}^{(s)}(E_g, 1) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_{2,2}^{(s)}(E_g, 1)}{2} \\
\text{z178} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, 2b) &= \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_{2,2}^{(s)}(E_g, 1) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_{2,1}^{(s)}(E_g, 1)}{2} \\
\text{z179} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 1) &= \frac{\sqrt{7}\left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{2,1}^{(s)}(E_g, 1) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{2,2}^{(s)}(E_g, 1) + 2\sqrt{3}\mathbb{Q}_2^{(1,-1;a)}(A_g)\mathbb{Q}_{2,1}^{(s)}(E_g, 1)\right)}{14} \\
\text{z180} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 1) &= \frac{\sqrt{7}\left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{2,2}^{(s)}(E_g, 1) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{2,1}^{(s)}(E_g, 1) + 2\sqrt{3}\mathbb{Q}_2^{(1,-1;a)}(A_g)\mathbb{Q}_{2,2}^{(s)}(E_g, 1)\right)}{14} \\
\text{z181} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, 1) &= -\frac{\sqrt{2}\mathbb{G}_1^{(1,0;a)}(A_g)\mathbb{Q}_{2,2}^{(s)}(E_g, 1)}{2} \\
\text{z182} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, 1) &= \frac{\sqrt{2}\mathbb{G}_1^{(1,0;a)}(A_g)\mathbb{Q}_{2,1}^{(s)}(E_g, 1)}{2} \\
\text{z183} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, 2) &= \frac{\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(E_g, 1) - \mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(E_g, 1)}{2}
\end{aligned}$$

$$\boxed{\text{z184}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, 2) = -\frac{\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(E_g, 1) + \mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(E_g, 1)}{2}$$

$$\boxed{\text{z185}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{2,1}^{(s)}(E_g, 1)}{2}$$

$$\boxed{\text{z186}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{2,2}^{(s)}(E_g, 1)}{2}$$

$$\boxed{\text{z187}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(E_g) = \frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_0^{(s)}(A_g)}{2}$$

$$\boxed{\text{z188}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(E_g) = \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_0^{(s)}(A_g)}{2}$$

• **A;A_001_1 : 'A'-'A' bond-cluster**

* bra: $\langle s, \uparrow |, \langle s, \downarrow |$

* ket: $|s, \uparrow \rangle, |s, \downarrow \rangle$

* wyckoff: **6a06g**

$$\boxed{\text{z17}} \quad \mathbb{Q}_0^{(c)}(A_g) = \mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z18}} \quad \mathbb{Q}_0^{(1,-1;c)}(A_g) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(E_g) + \mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z19}} \quad \mathbb{G}_1^{(1,-1;c)}(A_g, a) = \mathbb{M}_1^{(1,-1;a)}(A_g)\mathbb{T}_0^{(b)}(A_g)$$

$$\boxed{\text{z20}} \quad \mathbb{G}_1^{(1,-1;c)}(A_g, b) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(E_g) - \mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z93}} \quad \mathbb{Q}_1^{(c)}(A_u) = \mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_1^{(b)}(A_u)$$

$$\boxed{\text{z94}} \quad \mathbb{Q}_1^{(1,-1;c)}(A_u) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z95}} \quad \mathbb{G}_0^{(1,-1;c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_1^{(1,-1;a)}(A_g)\mathbb{T}_1^{(b)}(A_u) \right)}{3}$$

$$\boxed{\text{z96}} \quad \mathbb{G}_2^{(1,-1;c)}(A_u) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(E_u) + 2\mathbb{M}_1^{(1,-1;a)}(A_g) \mathbb{T}_1^{(b)}(A_u) \right)}{6}$$

$$\boxed{\text{z189}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, 1) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{2,1}^{(b)}(E_g, 1)}{2}$$

$$\boxed{\text{z190}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, 1) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{2,2}^{(b)}(E_g, 1)}{2}$$

$$\boxed{\text{z191}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, 1) = \frac{\sqrt{2} \mathbb{M}_1^{(1,-1;a)}(A_g) \mathbb{M}_{1,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z192}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, 1) = -\frac{\sqrt{2} \mathbb{M}_1^{(1,-1;a)}(A_g) \mathbb{M}_{1,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z193}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, 2) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E_g) \mathbb{M}_{1,1}^{(b)}(E_g) - \mathbb{M}_{1,2}^{(1,-1;a)}(E_g) \mathbb{M}_{1,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z194}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, 2) = -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E_g) \mathbb{M}_{1,2}^{(b)}(E_g) + \mathbb{M}_{1,2}^{(1,-1;a)}(E_g) \mathbb{M}_{1,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z195}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(E_g) = \frac{\sqrt{2} \mathbb{M}_{1,1}^{(1,-1;a)}(E_g) \mathbb{T}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z196}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(E_g) = \frac{\sqrt{2} \mathbb{M}_{1,2}^{(1,-1;a)}(E_g) \mathbb{T}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z341}} \quad \mathbb{Q}_{1,1}^{(c)}(E_u) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z342}} \quad \mathbb{Q}_{1,2}^{(c)}(E_u) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z343}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(E_u) = \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(E_g) \mathbb{T}_1^{(b)}(A_u) - \mathbb{M}_1^{(1,-1;a)}(A_g) \mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z344}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(E_u) = \frac{-\mathbb{M}_{1,1}^{(1,-1;a)}(E_g) \mathbb{T}_1^{(b)}(A_u) + \mathbb{M}_1^{(1,-1;a)}(A_g) \mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z345}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(Eu, 1) = \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(Eu)\mathbb{T}_1^{(b)}(Au) + \mathbb{M}_1^{(1,-1;a)}(Eu)\mathbb{T}_{1,2}^{(b)}(Eu)}{2}$$

$$\boxed{\text{z346}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(Eu, 1) = -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(Eu)\mathbb{T}_1^{(b)}(Au) + \mathbb{M}_1^{(1,-1;a)}(Eu)\mathbb{T}_{1,1}^{(b)}(Eu)}{2}$$

$$\boxed{\text{z347}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(Eu, 2) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(Eu)\mathbb{T}_{1,1}^{(b)}(Eu) - \mathbb{M}_{1,2}^{(1,-1;a)}(Eu)\mathbb{T}_{1,2}^{(b)}(Eu)}{2}$$

$$\boxed{\text{z348}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(Eu, 2) = -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(Eu)\mathbb{T}_{1,2}^{(b)}(Eu) + \mathbb{M}_{1,2}^{(1,-1;a)}(Eu)\mathbb{T}_{1,1}^{(b)}(Eu)}{2}$$

• **A;A_001_1** : 'A'-'A' bond-cluster

* bra: $\langle s, \uparrow |, \langle s, \downarrow |$

* ket: $|p_x, \uparrow\rangle, |p_x, \downarrow\rangle, |p_y, \uparrow\rangle, |p_y, \downarrow\rangle, |p_z, \uparrow\rangle, |p_z, \downarrow\rangle$

* wyckoff: **6a@6g**

$$\boxed{\text{z21}} \quad \mathbb{Q}_0^{(c)}(Ag, a) = \frac{\sqrt{3} \left(\mathbb{Q}_{1,1}^{(a)}(Eu)\mathbb{Q}_{1,1}^{(b)}(Eu) + \mathbb{Q}_{1,2}^{(a)}(Eu)\mathbb{Q}_{1,2}^{(b)}(Eu) + \mathbb{Q}_1^{(a)}(Au)\mathbb{Q}_1^{(b)}(Au) \right)}{3}$$

$$\boxed{\text{z22}} \quad \mathbb{Q}_0^{(c)}(Ag, b) = \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(a)}(Eu)\mathbb{T}_{1,1}^{(b)}(Eu) + \mathbb{T}_{1,2}^{(a)}(Eu)\mathbb{T}_{1,2}^{(b)}(Eu) + \mathbb{T}_1^{(a)}(Au)\mathbb{T}_1^{(b)}(Au) \right)}{3}$$

$$\boxed{\text{z23}} \quad \mathbb{Q}_2^{(c)}(Ag, a) = \frac{\sqrt{6} \left(-\mathbb{Q}_{1,1}^{(a)}(Eu)\mathbb{Q}_{1,1}^{(b)}(Eu) - \mathbb{Q}_{1,2}^{(a)}(Eu)\mathbb{Q}_{1,2}^{(b)}(Eu) + 2\mathbb{Q}_1^{(a)}(Au)\mathbb{Q}_1^{(b)}(Au) \right)}{6}$$

$$\boxed{\text{z24}} \quad \mathbb{Q}_2^{(c)}(Ag, b) = \frac{\sqrt{6} \left(-\mathbb{T}_{1,1}^{(a)}(Eu)\mathbb{T}_{1,1}^{(b)}(Eu) - \mathbb{T}_{1,2}^{(a)}(Eu)\mathbb{T}_{1,2}^{(b)}(Eu) + 2\mathbb{T}_1^{(a)}(Au)\mathbb{T}_1^{(b)}(Au) \right)}{6}$$

$$\boxed{\text{z25}} \quad \mathbb{Q}_2^{(1,-1;c)}(Ag, a) = -\frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(Eu, 1)\mathbb{Q}_{1,1}^{(b)}(Eu) + \mathbb{G}_{2,2}^{(1,-1;a)}(Eu, 1)\mathbb{Q}_{1,2}^{(b)}(Eu) \right)}{2}$$

$$\boxed{\text{z26}} \quad \mathbb{Q}_2^{(1,-1;c)}(Ag, b) = -\frac{\sqrt{2} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + \mathbb{M}_{2,2}^{(1,-1;a)}(Eu, 1)\mathbb{T}_{1,2}^{(b)}(Eu) \right)}{2}$$

$$\boxed{\text{z27}} \quad \mathbb{Q}_0^{(1,0;c)}(Ag, a) = \frac{\sqrt{3} \left(\mathbb{Q}_{1,1}^{(1,0;a)}(Eu)\mathbb{Q}_{1,1}^{(b)}(Eu) + \mathbb{Q}_{1,2}^{(1,0;a)}(Eu)\mathbb{Q}_{1,2}^{(b)}(Eu) + \mathbb{Q}_1^{(1,0;a)}(Au)\mathbb{Q}_1^{(b)}(Au) \right)}{3}$$

$$\begin{aligned}
\text{z28} \quad \mathbb{Q}_0^{(1,0;c)}(A_g, b) &= \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(1,0;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{1,2}^{(1,0;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{T}_1^{(1,0;a)}(A_u) \mathbb{T}_1^{(b)}(A_u) \right)}{3} \\
\text{z29} \quad \mathbb{Q}_2^{(1,0;c)}(A_g, a) &= \frac{\sqrt{6} \left(-\mathbb{Q}_{1,1}^{(1,0;a)}(E_u) \mathbb{Q}_{1,1}^{(b)}(E_u) - \mathbb{Q}_{1,2}^{(1,0;a)}(E_u) \mathbb{Q}_{1,2}^{(b)}(E_u) + 2\mathbb{Q}_1^{(1,0;a)}(A_u) \mathbb{Q}_1^{(b)}(A_u) \right)}{6} \\
\text{z30} \quad \mathbb{Q}_2^{(1,0;c)}(A_g, b) &= \frac{\sqrt{6} \left(-\mathbb{T}_{1,1}^{(1,0;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{T}_{1,2}^{(1,0;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(E_u) + 2\mathbb{T}_1^{(1,0;a)}(A_u) \mathbb{T}_1^{(b)}(A_u) \right)}{6} \\
\text{z31} \quad \mathbb{G}_1^{(c)}(A_g, a) &= \frac{\sqrt{2} \left(\mathbb{Q}_{1,1}^{(a)}(E_u) \mathbb{Q}_{1,2}^{(b)}(E_u) - \mathbb{Q}_{1,2}^{(a)}(E_u) \mathbb{Q}_{1,1}^{(b)}(E_u) \right)}{2} \\
\text{z32} \quad \mathbb{G}_1^{(c)}(A_g, b) &= \frac{\sqrt{2} \left(\mathbb{T}_{1,1}^{(a)}(E_u) \mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{T}_{1,2}^{(a)}(E_u) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2} \\
\text{z33} \quad \mathbb{G}_1^{(1,-1;c)}(A_g, a) &= \frac{\sqrt{10} \left(\sqrt{3} \mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1) \mathbb{Q}_{1,2}^{(b)}(E_u) - \sqrt{3} \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1) \mathbb{Q}_{1,1}^{(b)}(E_u) + 2\mathbb{G}_2^{(1,-1;a)}(A_u) \mathbb{Q}_1^{(b)}(A_u) \right)}{10} \\
\text{z34} \quad \mathbb{G}_1^{(1,-1;c)}(A_g, b) &= \frac{\sqrt{10} \left(\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3} \mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + 2\mathbb{M}_2^{(1,-1;a)}(A_u) \mathbb{T}_1^{(b)}(A_u) \right)}{10} \\
\text{z35} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, 1a) &= \frac{\sqrt{5} \left(-\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1) \mathbb{Q}_{1,2}^{(b)}(E_u) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1) \mathbb{Q}_{1,1}^{(b)}(E_u) + \sqrt{3} \mathbb{G}_2^{(1,-1;a)}(A_u) \mathbb{Q}_1^{(b)}(A_u) \right)}{5} \\
\text{z36} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, 1b) &= \frac{\sqrt{5} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3} \mathbb{M}_2^{(1,-1;a)}(A_u) \mathbb{T}_1^{(b)}(A_u) \right)}{5} \\
\text{z37} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, 2a) &= \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2) \mathbb{Q}_{1,2}^{(b)}(E_u) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2) \mathbb{Q}_{1,1}^{(b)}(E_u) \right)}{2} \\
\text{z38} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, 2b) &= \frac{\sqrt{2} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 2) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2} \\
\text{z39} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, 3a) &= \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2) \mathbb{Q}_{1,1}^{(b)}(E_u) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2) \mathbb{Q}_{1,2}^{(b)}(E_u) \right)}{2} \\
\text{z40} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, 3b) &= \frac{\sqrt{2} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 2) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2}
\end{aligned}$$

$$\boxed{\text{z41}} \quad \mathbb{G}_1^{(1,0;c)}(A_g, a) = \frac{\sqrt{2} \left(\mathbb{Q}_{1,1}^{(1,0;a)}(E_u) \mathbb{Q}_{1,2}^{(b)}(E_u) - \mathbb{Q}_{1,2}^{(1,0;a)}(E_u) \mathbb{Q}_{1,1}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z42}} \quad \mathbb{G}_1^{(1,0;c)}(A_g, b) = \frac{\sqrt{2} \left(\mathbb{T}_{1,1}^{(1,0;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{T}_{1,2}^{(1,0;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z43}} \quad \mathbb{G}_1^{(1,1;c)}(A_g, a) = \mathbb{G}_0^{(1,1;a)}(A_u) \mathbb{Q}_1^{(b)}(A_u)$$

$$\boxed{\text{z44}} \quad \mathbb{G}_1^{(1,1;c)}(A_g, b) = \mathbb{M}_0^{(1,1;a)}(A_u) \mathbb{T}_1^{(b)}(A_u)$$

$$\boxed{\text{z97}} \quad \mathbb{Q}_1^{(c)}(A_u, a) = \mathbb{Q}_1^{(a)}(A_u) \mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z98}} \quad \mathbb{Q}_1^{(c)}(A_u, b) = \frac{\sqrt{2} \left(-\mathbb{Q}_{1,1}^{(a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g, 1) + \mathbb{Q}_{1,2}^{(a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z99}} \quad \mathbb{Q}_1^{(c)}(A_u, c) = \mathbb{T}_1^{(a)}(A_u) \mathbb{T}_0^{(b)}(A_g)$$

$$\boxed{\text{z100}} \quad \mathbb{Q}_1^{(c)}(A_u, d) = \frac{\sqrt{2} \left(\mathbb{T}_{1,1}^{(a)}(E_u) \mathbb{M}_{1,2}^{(b)}(E_g) - \mathbb{T}_{1,2}^{(a)}(E_u) \mathbb{M}_{1,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z101}} \quad \mathbb{Q}_1^{(1,-1;c)}(A_u, a) = \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1) \mathbb{Q}_{2,2}^{(b)}(E_g, 1) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1) \mathbb{Q}_{2,1}^{(b)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z102}} \quad \mathbb{Q}_1^{(1,-1;c)}(A_u, b) = \frac{\sqrt{2} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 1) \mathbb{M}_{1,2}^{(b)}(E_g) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 1) \mathbb{M}_{1,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z103}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, 2a) = \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2) \mathbb{Q}_{2,2}^{(b)}(E_g, 1) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2) \mathbb{Q}_{2,1}^{(b)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z104}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, 2b) = \frac{\sqrt{2} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 2) \mathbb{M}_{1,2}^{(b)}(E_g) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 2) \mathbb{M}_{1,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z105}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, 3a) = \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2) \mathbb{Q}_{2,1}^{(b)}(E_g, 1) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2) \mathbb{Q}_{2,2}^{(b)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z106}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, 3b) = \frac{\sqrt{2} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 2) \mathbb{M}_{1,1}^{(b)}(E_g) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 2) \mathbb{M}_{1,2}^{(b)}(E_g) \right)}{2}$$

$$\begin{aligned}
\boxed{\text{z107}} \quad \mathbb{Q}_1^{(1,0;c)}(A_u, a) &= \mathbb{Q}_1^{(1,0;a)}(A_u) \mathbb{Q}_0^{(b)}(A_g) \\
\boxed{\text{z108}} \quad \mathbb{Q}_1^{(1,0;c)}(A_u, b) &= \frac{\sqrt{2} \left(-\mathbb{Q}_{1,1}^{(1,0;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g, 1) + \mathbb{Q}_{1,2}^{(1,0;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g, 1) \right)}{2} \\
\boxed{\text{z109}} \quad \mathbb{Q}_1^{(1,0;c)}(A_u, c) &= \mathbb{T}_1^{(1,0;a)}(A_u) \mathbb{T}_0^{(b)}(A_g) \\
\boxed{\text{z110}} \quad \mathbb{Q}_1^{(1,0;c)}(A_u, d) &= \frac{\sqrt{2} \left(\mathbb{T}_{1,1}^{(1,0;a)}(E_u) \mathbb{M}_{1,2}^{(b)}(E_g) - \mathbb{T}_{1,2}^{(1,0;a)}(E_u) \mathbb{M}_{1,1}^{(b)}(E_g) \right)}{2} \\
\boxed{\text{z111}} \quad \mathbb{G}_0^{(c)}(A_u) &= \frac{\sqrt{2} \left(\mathbb{T}_{1,1}^{(a)}(E_u) \mathbb{M}_{1,1}^{(b)}(E_g) + \mathbb{T}_{1,2}^{(a)}(E_u) \mathbb{M}_{1,2}^{(b)}(E_g) \right)}{2} \\
\boxed{\text{z112}} \quad \mathbb{G}_2^{(c)}(A_u) &= \frac{\sqrt{2} \left(\mathbb{Q}_{1,1}^{(a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g, 1) + \mathbb{Q}_{1,2}^{(a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g, 1) \right)}{2} \\
\boxed{\text{z113}} \quad \mathbb{G}_0^{(1,-1;c)}(A_u) &= \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1) \mathbb{Q}_{2,1}^{(b)}(E_g, 1) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1) \mathbb{Q}_{2,2}^{(b)}(E_g, 1) \right)}{2} \\
\boxed{\text{z114}} \quad \mathbb{G}_2^{(1,-1;c)}(A_u, a) &= \mathbb{G}_2^{(1,-1;a)}(A_u) \mathbb{Q}_0^{(b)}(A_g) \\
\boxed{\text{z115}} \quad \mathbb{G}_2^{(1,-1;c)}(A_u, b) &= \mathbb{M}_2^{(1,-1;a)}(A_u) \mathbb{T}_0^{(b)}(A_g) \\
\boxed{\text{z116}} \quad \mathbb{G}_2^{(1,-1;c)}(A_u, c) &= -\frac{\sqrt{2} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 1) \mathbb{M}_{1,1}^{(b)}(E_g) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 1) \mathbb{M}_{1,2}^{(b)}(E_g) \right)}{2} \\
\boxed{\text{z117}} \quad \mathbb{G}_0^{(1,0;c)}(A_u) &= \frac{\sqrt{2} \left(\mathbb{T}_{1,1}^{(1,0;a)}(E_u) \mathbb{M}_{1,1}^{(b)}(E_g) + \mathbb{T}_{1,2}^{(1,0;a)}(E_u) \mathbb{M}_{1,2}^{(b)}(E_g) \right)}{2} \\
\boxed{\text{z118}} \quad \mathbb{G}_2^{(1,0;c)}(A_u) &= \frac{\sqrt{2} \left(\mathbb{Q}_{1,1}^{(1,0;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g, 1) + \mathbb{Q}_{1,2}^{(1,0;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g, 1) \right)}{2} \\
\boxed{\text{z119}} \quad \mathbb{G}_0^{(1,1;c)}(A_u, a) &= \mathbb{G}_0^{(1,1;a)}(A_u) \mathbb{Q}_0^{(b)}(A_g) \\
\boxed{\text{z120}} \quad \mathbb{G}_0^{(1,1;c)}(A_u, b) &= \mathbb{M}_0^{(1,1;a)}(A_u) \mathbb{T}_0^{(b)}(A_g) \\
\boxed{\text{z197}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, 1a) &= \frac{\mathbb{Q}_{1,2}^{(a)}(E_u) \mathbb{Q}_1^{(b)}(A_u) + \mathbb{Q}_1^{(a)}(A_u) \mathbb{Q}_{1,2}^{(b)}(E_u)}{2}
\end{aligned}$$

$$\boxed{\text{z198}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, 1a) = -\frac{\mathbb{Q}_{1,1}^{(a)}(E_u)\mathbb{Q}_1^{(b)}(A_u) + \mathbb{Q}_1^{(a)}(A_u)\mathbb{Q}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z199}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, 1b) = \frac{\mathbb{T}_{1,2}^{(a)}(E_u)\mathbb{T}_1^{(b)}(A_u) + \mathbb{T}_1^{(a)}(A_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z200}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, 1b) = -\frac{\mathbb{T}_{1,1}^{(a)}(E_u)\mathbb{T}_1^{(b)}(A_u) + \mathbb{T}_1^{(a)}(A_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z201}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, 2a) = \frac{\mathbb{Q}_{1,1}^{(a)}(E_u)\mathbb{Q}_{1,1}^{(b)}(E_u) - \mathbb{Q}_{1,2}^{(a)}(E_u)\mathbb{Q}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z202}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, 2a) = -\frac{\mathbb{Q}_{1,1}^{(a)}(E_u)\mathbb{Q}_{1,2}^{(b)}(E_u) + \mathbb{Q}_{1,2}^{(a)}(E_u)\mathbb{Q}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z203}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, 2b) = \frac{\mathbb{T}_{1,1}^{(a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{T}_{1,2}^{(a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z204}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, 2b) = -\frac{\mathbb{T}_{1,1}^{(a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{T}_{1,2}^{(a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z205}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, 1a) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_1^{(b)}(A_u) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{1,2}^{(b)}(E_u) + 3\mathbb{G}_2^{(1,-1;a)}(A_u)\mathbb{Q}_{1,1}^{(b)}(E_u)}{6}$$

$$\boxed{\text{z206}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, 1a) = \frac{-\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_1^{(b)}(A_u) - \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{1,1}^{(b)}(E_u) + 3\mathbb{G}_2^{(1,-1;a)}(A_u)\mathbb{Q}_{1,2}^{(b)}(E_u)}{6}$$

$$\boxed{\text{z207}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, 1b) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_1^{(b)}(A_u) - \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 3\mathbb{M}_2^{(1,-1;a)}(A_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{6}$$

$$\boxed{\text{z208}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, 1b) = \frac{-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_1^{(b)}(A_u) - \sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\mathbb{M}_2^{(1,-1;a)}(A_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{6}$$

$$\boxed{\text{z209}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, 2a) = \frac{\sqrt{3} \left(-\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{1,1}^{(b)}(E_u) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{1,2}^{(b)}(E_u) - 2\mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_1^{(b)}(A_u) \right)}{6}$$

$$\boxed{\text{z210}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, 2a) = \frac{\sqrt{3} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{1,2}^{(b)}(E_u) + 2\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_1^{(b)}(A_u) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{1,1}^{(b)}(E_u) \right)}{6}$$

$$\begin{aligned}
\text{z211} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, 2b) &= \frac{\sqrt{3} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - 2\mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 2) \mathbb{T}_1^{(b)}(A_u) \right)}{6} \\
\text{z212} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, 2b) &= \frac{\sqrt{3} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + 2\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 2) \mathbb{T}_1^{(b)}(A_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 1) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{6} \\
\text{z213} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, 1a) &= \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(E_u) \mathbb{Q}_1^{(b)}(A_u) + \mathbb{Q}_1^{(1,0;a)}(A_u) \mathbb{Q}_{1,2}^{(b)}(E_u)}{2} \\
\text{z214} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, 1a) &= -\frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E_u) \mathbb{Q}_1^{(b)}(A_u) + \mathbb{Q}_1^{(1,0;a)}(A_u) \mathbb{Q}_{1,1}^{(b)}(E_u)}{2} \\
\text{z215} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, 1b) &= \frac{\mathbb{T}_{1,2}^{(1,0;a)}(E_u) \mathbb{T}_1^{(b)}(A_u) + \mathbb{T}_1^{(1,0;a)}(A_u) \mathbb{T}_{1,2}^{(b)}(E_u)}{2} \\
\text{z216} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, 1b) &= -\frac{\mathbb{T}_{1,1}^{(1,0;a)}(E_u) \mathbb{T}_1^{(b)}(A_u) + \mathbb{T}_1^{(1,0;a)}(A_u) \mathbb{T}_{1,1}^{(b)}(E_u)}{2} \\
\text{z217} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, 2a) &= \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E_u) \mathbb{Q}_{1,1}^{(b)}(E_u) - \mathbb{Q}_{1,2}^{(1,0;a)}(E_u) \mathbb{Q}_{1,2}^{(b)}(E_u)}{2} \\
\text{z218} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, 2a) &= -\frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E_u) \mathbb{Q}_{1,2}^{(b)}(E_u) + \mathbb{Q}_{1,2}^{(1,0;a)}(E_u) \mathbb{Q}_{1,1}^{(b)}(E_u)}{2} \\
\text{z219} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, 2b) &= \frac{\mathbb{T}_{1,1}^{(1,0;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{T}_{1,2}^{(1,0;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(E_u)}{2} \\
\text{z220} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, 2b) &= -\frac{\mathbb{T}_{1,1}^{(1,0;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{T}_{1,2}^{(1,0;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(E_u)}{2} \\
\text{z221} \quad \mathbb{G}_{1,1}^{(c)}(E_g, a) &= \frac{\mathbb{Q}_{1,2}^{(a)}(E_u) \mathbb{Q}_1^{(b)}(A_u) - \mathbb{Q}_1^{(a)}(A_u) \mathbb{Q}_{1,2}^{(b)}(E_u)}{2} \\
\text{z222} \quad \mathbb{G}_{1,2}^{(c)}(E_g, a) &= \frac{-\mathbb{Q}_{1,1}^{(a)}(E_u) \mathbb{Q}_1^{(b)}(A_u) + \mathbb{Q}_1^{(a)}(A_u) \mathbb{Q}_{1,1}^{(b)}(E_u)}{2} \\
\text{z223} \quad \mathbb{G}_{1,1}^{(c)}(E_g, b) &= \frac{\mathbb{T}_{1,2}^{(a)}(E_u) \mathbb{T}_1^{(b)}(A_u) - \mathbb{T}_1^{(a)}(A_u) \mathbb{T}_{1,2}^{(b)}(E_u)}{2}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z224}} \quad \mathbb{G}_{1,2}^{(c)}(E_g, b) &= \frac{-\mathbb{T}_{1,1}^{(a)}(E_u)\mathbb{T}_1^{(b)}(A_u) + \mathbb{T}_1^{(a)}(A_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} \\
\boxed{\text{z225}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(E_g, a) &= \frac{\sqrt{5} \left(\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_1^{(b)}(A_u) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{1,2}^{(b)}(E_u) - \mathbb{G}_2^{(1,-1;a)}(A_u)\mathbb{Q}_{1,1}^{(b)}(E_u) \right)}{10} \\
\boxed{\text{z226}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(E_g, a) &= \frac{\sqrt{5} \left(\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_1^{(b)}(A_u) - \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{1,1}^{(b)}(E_u) - \mathbb{G}_2^{(1,-1;a)}(A_u)\mathbb{Q}_{1,2}^{(b)}(E_u) \right)}{10} \\
\boxed{\text{z227}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(E_g, b) &= \frac{\sqrt{5} \left(\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_1^{(b)}(A_u) - \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_2^{(1,-1;a)}(A_u)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{10} \\
\boxed{\text{z228}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(E_g, b) &= \frac{\sqrt{5} \left(\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_1^{(b)}(A_u) - \sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_2^{(1,-1;a)}(A_u)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{10} \\
\boxed{\text{z229}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(E_g, 1a) &= \frac{\sqrt{5} \left(-\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{1,1}^{(b)}(E_u) - 4\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_1^{(b)}(A_u) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{1,2}^{(b)}(E_u) + 6\mathbb{G}_2^{(1,-1;a)}(A_u)\mathbb{Q}_{1,1}^{(b)}(E_u) \right)}{30} \\
\boxed{\text{z230}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(E_g, 1a) &= \frac{\sqrt{5} \left(4\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_1^{(b)}(A_u) + \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{1,1}^{(b)}(E_u) + 6\mathbb{G}_2^{(1,-1;a)}(A_u)\mathbb{Q}_{1,2}^{(b)}(E_u) \right)}{30} \\
\boxed{\text{z231}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(E_g, 1b) &= \frac{\sqrt{5} \left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 4\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_1^{(b)}(A_u) + \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 6\mathbb{M}_2^{(1,-1;a)}(A_u)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{30} \\
\boxed{\text{z232}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(E_g, 1b) &= \frac{\sqrt{5} \left(4\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_1^{(b)}(A_u) + \sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 6\mathbb{M}_2^{(1,-1;a)}(A_u)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{30} \\
\boxed{\text{z233}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(E_g, 2a) &= \frac{\sqrt{6} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{1,1}^{(b)}(E_u) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{1,2}^{(b)}(E_u) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_1^{(b)}(A_u) \right)}{6} \\
\boxed{\text{z234}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(E_g, 2a) &= \frac{\sqrt{6} \left(-\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{1,2}^{(b)}(E_u) + \mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_1^{(b)}(A_u) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{1,1}^{(b)}(E_u) \right)}{6} \\
\boxed{\text{z235}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(E_g, 2b) &= \frac{\sqrt{6} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_1^{(b)}(A_u) \right)}{6} \\
\boxed{\text{z236}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(E_g, 2b) &= \frac{\sqrt{6} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_1^{(b)}(A_u) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{6}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z237}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(E_g, a) &= \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(E_u)\mathbb{Q}_1^{(b)}(A_u) - \mathbb{Q}_1^{(1,0;a)}(A_u)\mathbb{Q}_{1,2}^{(b)}(E_u)}{2} \\
\boxed{\text{z238}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(E_g, a) &= \frac{-\mathbb{Q}_{1,1}^{(1,0;a)}(E_u)\mathbb{Q}_1^{(b)}(A_u) + \mathbb{Q}_1^{(1,0;a)}(A_u)\mathbb{Q}_{1,1}^{(b)}(E_u)}{2} \\
\boxed{\text{z239}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(E_g, b) &= \frac{\mathbb{T}_{1,2}^{(1,0;a)}(E_u)\mathbb{T}_1^{(b)}(A_u) - \mathbb{T}_1^{(1,0;a)}(A_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} \\
\boxed{\text{z240}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(E_g, b) &= \frac{-\mathbb{T}_{1,1}^{(1,0;a)}(E_u)\mathbb{T}_1^{(b)}(A_u) + \mathbb{T}_1^{(1,0;a)}(A_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} \\
\boxed{\text{z241}} \quad \mathbb{G}_{1,1}^{(1,1;c)}(E_g, a) &= \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{1,1}^{(b)}(E_u)}{2} \\
\boxed{\text{z242}} \quad \mathbb{G}_{1,2}^{(1,1;c)}(E_g, a) &= \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{1,2}^{(b)}(E_u)}{2} \\
\boxed{\text{z243}} \quad \mathbb{G}_{1,1}^{(1,1;c)}(E_g, b) &= \frac{\sqrt{2}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} \\
\boxed{\text{z244}} \quad \mathbb{G}_{1,2}^{(1,1;c)}(E_g, b) &= \frac{\sqrt{2}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} \\
\boxed{\text{z349}} \quad \mathbb{Q}_{1,1}^{(c)}(E_u, a) &= \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(E_u)\mathbb{Q}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z350}} \quad \mathbb{Q}_{1,2}^{(c)}(E_u, a) &= \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(E_u)\mathbb{Q}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z351}} \quad \mathbb{Q}_{1,1}^{(c)}(E_u, b) &= -\frac{\sqrt{2}\mathbb{Q}_1^{(a)}(A_u)\mathbb{Q}_{2,2}^{(b)}(E_g, 1)}{2} \\
\boxed{\text{z352}} \quad \mathbb{Q}_{1,2}^{(c)}(E_u, b) &= \frac{\sqrt{2}\mathbb{Q}_1^{(a)}(A_u)\mathbb{Q}_{2,1}^{(b)}(E_g, 1)}{2} \\
\boxed{\text{z353}} \quad \mathbb{Q}_{1,1}^{(c)}(E_u, c) &= \frac{\sqrt{2}\mathbb{T}_{1,1}^{(a)}(E_u)\mathbb{T}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z354}} \quad \mathbb{Q}_{1,2}^{(c)}(E_u, c) &= \frac{\sqrt{2}\mathbb{T}_{1,2}^{(a)}(E_u)\mathbb{T}_0^{(b)}(A_g)}{2}
\end{aligned}$$

$$\begin{aligned}
\text{z355} \quad \mathbb{Q}_{1,1}^{(c)}(E_u, d) &= -\frac{\sqrt{2}\mathbb{T}_1^{(a)}(A_u)\mathbb{M}_{1,2}^{(b)}(E_g)}{2} \\
\text{z356} \quad \mathbb{Q}_{1,2}^{(c)}(E_u, d) &= \frac{\sqrt{2}\mathbb{T}_1^{(a)}(A_u)\mathbb{M}_{1,1}^{(b)}(E_g)}{2} \\
\text{z357} \quad \mathbb{Q}_{3,1}^{(c)}(E_u, 2) &= \frac{\mathbb{Q}_{1,1}^{(a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) - \mathbb{Q}_{1,2}^{(a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g, 1)}{2} \\
\text{z358} \quad \mathbb{Q}_{3,2}^{(c)}(E_u, 2) &= -\frac{\mathbb{Q}_{1,1}^{(a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) + \mathbb{Q}_{1,2}^{(a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g, 1)}{2} \\
\text{z359} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(E_u, a) &= \frac{\sqrt{10} \left(-\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) - \sqrt{3}\mathbb{G}_2^{(1,-1;a)}(A_u)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) \right)}{10} \\
\text{z360} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(E_u, a) &= \frac{\sqrt{10} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) - \sqrt{3}\mathbb{G}_2^{(1,-1;a)}(A_u)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) \right)}{10} \\
\text{z361} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(E_u, b) &= \frac{\sqrt{14} \left(\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{M}_{1,1}^{(b)}(E_g) - \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{M}_{1,2}^{(b)}(E_g) - \mathbb{M}_2^{(1,-1;a)}(A_u)\mathbb{M}_{1,1}^{(b)}(E_g) \right)}{14} \\
\text{z362} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(E_u, b) &= -\frac{\sqrt{14} \left(\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{M}_{1,2}^{(b)}(E_g) + \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{M}_{1,1}^{(b)}(E_g) + \mathbb{M}_2^{(1,-1;a)}(A_u)\mathbb{M}_{1,2}^{(b)}(E_g) \right)}{14} \\
\text{z363} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 1a) &= \frac{\sqrt{5} \left(\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) - 2\mathbb{G}_2^{(1,-1;a)}(A_u)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) \right)}{10} \\
\text{z364} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 1a) &= -\frac{\sqrt{5} \left(\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) + 2\mathbb{G}_2^{(1,-1;a)}(A_u)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) \right)}{10} \\
\text{z365} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 1b) &= \frac{\sqrt{7} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{M}_{1,1}^{(b)}(E_g) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{M}_{1,2}^{(b)}(E_g) + 2\sqrt{3}\mathbb{M}_2^{(1,-1;a)}(A_u)\mathbb{M}_{1,1}^{(b)}(E_g) \right)}{14} \\
\text{z366} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 1b) &= \frac{\sqrt{7} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{M}_{1,2}^{(b)}(E_g) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{M}_{1,1}^{(b)}(E_g) + 2\sqrt{3}\mathbb{M}_2^{(1,-1;a)}(A_u)\mathbb{M}_{1,2}^{(b)}(E_g) \right)}{14} \\
\text{z367} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 2) &= \frac{\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{M}_{1,1}^{(b)}(E_g) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{M}_{1,2}^{(b)}(E_g)}{2}
\end{aligned}$$

$$\boxed{\text{z368}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 2) = -\frac{\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{M}_{1,2}^{(b)}(E_g) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{M}_{1,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z369}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(E_u, a) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(E_u)\mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z370}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(E_u, a) = \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(E_u)\mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z371}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(E_u, b) = -\frac{\sqrt{2}\mathbb{Q}_1^{(1,0;a)}(A_u)\mathbb{Q}_{2,2}^{(b)}(E_g, 1)}{2}$$

$$\boxed{\text{z372}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(E_u, b) = \frac{\sqrt{2}\mathbb{Q}_1^{(1,0;a)}(A_u)\mathbb{Q}_{2,1}^{(b)}(E_g, 1)}{2}$$

$$\boxed{\text{z373}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(E_u, c) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(1,0;a)}(E_u)\mathbb{T}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z374}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(E_u, c) = \frac{\sqrt{2}\mathbb{T}_{1,2}^{(1,0;a)}(E_u)\mathbb{T}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z375}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(E_u, d) = -\frac{\sqrt{2}\mathbb{T}_1^{(1,0;a)}(A_u)\mathbb{M}_{1,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z376}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(E_u, d) = \frac{\sqrt{2}\mathbb{T}_1^{(1,0;a)}(A_u)\mathbb{M}_{1,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z377}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 2) = \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) - \mathbb{Q}_{1,2}^{(1,0;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g, 1)}{2}$$

$$\boxed{\text{z378}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 2) = -\frac{\mathbb{Q}_{1,1}^{(1,0;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) + \mathbb{Q}_{1,2}^{(1,0;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g, 1)}{2}$$

$$\boxed{\text{z379}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{M}_{1,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z380}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{M}_{1,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z381}} \quad \mathbb{G}_{2,1}^{(c)}(E_u, 2) = \frac{\mathbb{T}_{1,1}^{(a)}(E_u)\mathbb{M}_{1,1}^{(b)}(E_g) - \mathbb{T}_{1,2}^{(a)}(E_u)\mathbb{M}_{1,2}^{(b)}(E_g)}{2}$$

$$\begin{aligned}
\boxed{\text{z382}} \quad \mathbb{G}_{2,2}^{(c)}(E_u, 2) &= -\frac{\mathbb{T}_{1,1}^{(a)}(E_u)\mathbb{M}_{1,2}^{(b)}(E_g) + \mathbb{T}_{1,2}^{(a)}(E_u)\mathbb{M}_{1,1}^{(b)}(E_g)}{2} \\
\boxed{\text{z383}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, 1a) &= \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z384}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, 1a) &= \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z385}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, 1b) &= \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{T}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z386}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, 1b) &= \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{T}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z387}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, 2a) &= \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z388}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, 2a) &= \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{Q}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z389}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, 2b) &= \frac{-\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{2,2}^{(b)}(E_g, 1)}{2} \\
\boxed{\text{z390}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, 2b) &= \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1)\mathbb{Q}_{2,1}^{(b)}(E_g, 1)}{2} \\
\boxed{\text{z391}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, 2c) &= \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 2)\mathbb{T}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z392}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, 2c) &= \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 2)\mathbb{T}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z393}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u, 2) &= \frac{\mathbb{T}_{1,1}^{(1,0;a)}(E_u)\mathbb{M}_{1,1}^{(b)}(E_g) - \mathbb{T}_{1,2}^{(1,0;a)}(E_u)\mathbb{M}_{1,2}^{(b)}(E_g)}{2} \\
\boxed{\text{z394}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u, 2) &= -\frac{\mathbb{T}_{1,1}^{(1,0;a)}(E_u)\mathbb{M}_{1,2}^{(b)}(E_g) + \mathbb{T}_{1,2}^{(1,0;a)}(E_u)\mathbb{M}_{1,1}^{(b)}(E_g)}{2} \\
\boxed{\text{z395}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u, 1) &= \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{2,1}^{(b)}(E_g, 1)}{2}
\end{aligned}$$

$$\boxed{\text{z396}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u, 1) = \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{2,2}^{(b)}(E_g, 1)}{2}$$

• **A;A_001_1** : 'A'-'A' bond-cluster

* bra: $\langle p_x, \uparrow |, \langle p_x, \downarrow |, \langle p_y, \uparrow |, \langle p_y, \downarrow |, \langle p_z, \uparrow |, \langle p_z, \downarrow |$

* ket: $|p_x, \uparrow \rangle, |p_x, \downarrow \rangle, |p_y, \uparrow \rangle, |p_y, \downarrow \rangle, |p_z, \uparrow \rangle, |p_z, \downarrow \rangle$

* wyckoff: 6a@6g

$$\boxed{\text{z45}} \quad \mathbb{Q}_0^{(c)}(A_g, a) = \mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z46}} \quad \mathbb{Q}_0^{(c)}(A_g, b) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g, 1)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) + \mathbb{Q}_{2,2}^{(a)}(E_g, 1)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z47}} \quad \mathbb{Q}_0^{(c)}(A_g, c) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{M}_{1,1}^{(b)}(E_g) + \mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{M}_{1,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z48}} \quad \mathbb{Q}_2^{(c)}(A_g) = \mathbb{Q}_2^{(a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z49}} \quad \mathbb{Q}_4^{(c)}(A_g, 2) = \frac{\sqrt{2} \left(-\mathbb{Q}_{2,1}^{(a)}(E_g, 2)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) + \mathbb{Q}_{2,2}^{(a)}(E_g, 2)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z50}} \quad \mathbb{Q}_4^{(c)}(A_g, 3) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g, 2)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) + \mathbb{Q}_{2,2}^{(a)}(E_g, 2)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z51}} \quad \mathbb{Q}_0^{(1,-1;c)}(A_g, a) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z52}} \quad \mathbb{Q}_0^{(1,-1;c)}(A_g, b) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(E_g) + \mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z53}} \quad \mathbb{Q}_2^{(1,-1;c)}(A_g, a) = \mathbb{Q}_2^{(1,-1;a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z54}} \quad \mathbb{Q}_2^{(1,-1;c)}(A_g, b) = \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{M}_{1,1}^{(b)}(E_g) + \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{M}_{1,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z55}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_g, 2a) = \frac{\sqrt{2} \left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z56}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_g, 2b) = \frac{\sqrt{2} \left(-\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2) \mathbb{M}_{1,2}^{(b)}(E_g) + \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2) \mathbb{M}_{1,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z57}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_g, 3a) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2) \mathbb{Q}_{2,1}^{(b)}(E_g, 1) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2) \mathbb{Q}_{2,2}^{(b)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z58}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_g, 3b) = \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2) \mathbb{M}_{1,1}^{(b)}(E_g) + \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2) \mathbb{M}_{1,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z59}} \quad \mathbb{Q}_2^{(1,0;c)}(A_g, a) = \frac{\sqrt{2} \left(\mathbb{G}_{1,1}^{(1,0;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g, 1) + \mathbb{G}_{1,2}^{(1,0;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z60}} \quad \mathbb{Q}_2^{(1,0;c)}(A_g, b) = \mathbb{T}_2^{(1,0;a)}(A_g) \mathbb{T}_0^{(b)}(A_g)$$

$$\boxed{\text{z61}} \quad \mathbb{Q}_2^{(1,0;c)}(A_g, c) = -\frac{\sqrt{2} \left(\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 1) \mathbb{M}_{1,1}^{(b)}(E_g) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 1) \mathbb{M}_{1,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z62}} \quad \mathbb{Q}_0^{(1,1;c)}(A_g, a) = \mathbb{Q}_0^{(1,1;a)}(A_g) \mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z63}} \quad \mathbb{Q}_0^{(1,1;c)}(A_g, b) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,1;a)}(E_g) \mathbb{M}_{1,1}^{(b)}(E_g) + \mathbb{M}_{1,2}^{(1,1;a)}(E_g) \mathbb{M}_{1,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z64}} \quad \mathbb{G}_1^{(c)}(A_g, a) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g, 1) \mathbb{Q}_{2,2}^{(b)}(E_g, 1) - \mathbb{Q}_{2,2}^{(a)}(E_g, 1) \mathbb{Q}_{2,1}^{(b)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z65}} \quad \mathbb{G}_1^{(c)}(A_g, b) = \mathbb{M}_1^{(a)}(A_g) \mathbb{T}_0^{(b)}(A_g)$$

$$\boxed{\text{z66}} \quad \mathbb{G}_1^{(c)}(A_g, c) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_g) \mathbb{M}_{1,2}^{(b)}(E_g) - \mathbb{M}_{1,2}^{(a)}(E_g) \mathbb{M}_{1,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z67}} \quad \mathbb{G}_1^{(1,-1;c)}(A_g, a) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1) \mathbb{Q}_{2,2}^{(b)}(E_g, 1) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1) \mathbb{Q}_{2,1}^{(b)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z68}} \quad \mathbb{G}_1^{(1,-1;c)}(A_g, b) = \mathbb{M}_1^{(1,-1;a)}(A_g) \mathbb{T}_0^{(b)}(A_g)$$

$$\boxed{\text{z69}} \quad \mathbb{G}_1^{(1,-1;c)}(A_g, c) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(E_g) \mathbb{M}_{1,2}^{(b)}(E_g) - \mathbb{M}_{1,2}^{(1,-1;a)}(E_g) \mathbb{M}_{1,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z70}} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, 1a) = \mathbb{M}_3^{(1,-1;a)}(A_g, 1)\mathbb{T}_0^{(b)}(A_g)$$

$$\boxed{\text{z71}} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, 1b) = \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{M}_{1,2}^{(b)}(E_g) - \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{M}_{1,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z72}} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, 2) = \mathbb{M}_3^{(1,-1;a)}(A_g, 2)\mathbb{T}_0^{(b)}(A_g)$$

$$\boxed{\text{z73}} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, 3) = \mathbb{M}_3^{(1,-1;a)}(A_g, 3)\mathbb{T}_0^{(b)}(A_g)$$

$$\boxed{\text{z74}} \quad \mathbb{G}_1^{(1,0;c)}(A_g, a) = \mathbb{G}_1^{(1,0;a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z75}} \quad \mathbb{G}_1^{(1,0;c)}(A_g, b) = \frac{\sqrt{2} \left(-\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) + \mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) \right)}{2}$$

$$\boxed{\text{z76}} \quad \mathbb{G}_1^{(1,0;c)}(A_g, c) = \frac{\sqrt{2} \left(\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 1)\mathbb{M}_{1,2}^{(b)}(E_g) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 1)\mathbb{M}_{1,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z77}} \quad \mathbb{G}_3^{(1,0;c)}(A_g, 2) = \frac{\sqrt{2} \left(\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 2)\mathbb{M}_{1,2}^{(b)}(E_g) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 2)\mathbb{M}_{1,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z78}} \quad \mathbb{G}_3^{(1,0;c)}(A_g, 3) = \frac{\sqrt{2} \left(\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 2)\mathbb{M}_{1,1}^{(b)}(E_g) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 2)\mathbb{M}_{1,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z79}} \quad \mathbb{G}_1^{(1,1;c)}(A_g, a) = \mathbb{M}_1^{(1,1;a)}(A_g)\mathbb{T}_0^{(b)}(A_g)$$

$$\boxed{\text{z80}} \quad \mathbb{G}_1^{(1,1;c)}(A_g, b) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(E_g) - \mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z121}} \quad \mathbb{Q}_1^{(c)}(A_u, a) = \mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_1^{(b)}(A_u)$$

$$\boxed{\text{z122}} \quad \mathbb{Q}_1^{(c)}(A_u, b) = \frac{\sqrt{10} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g, 1)\mathbb{Q}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g, 1)\mathbb{Q}_{1,1}^{(b)}(E_u) + 2\mathbb{Q}_2^{(a)}(A_g)\mathbb{Q}_1^{(b)}(A_u) \right)}{10}$$

$$\boxed{\text{z123}} \quad \mathbb{Q}_1^{(c)}(A_u, c) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z124}} \quad \mathbb{Q}_3^{(c)}(A_u, 1) = \frac{\sqrt{5} \left(-\mathbb{Q}_{2,1}^{(a)}(E_g, 1)\mathbb{Q}_{1,2}^{(b)}(E_u) + \mathbb{Q}_{2,2}^{(a)}(E_g, 1)\mathbb{Q}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{Q}_2^{(a)}(A_g)\mathbb{Q}_1^{(b)}(A_u) \right)}{5}$$

$$\begin{aligned}
\text{z125} \quad \mathbb{Q}_3^{(c)}(A_u, 2) &= \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g, 2) \mathbb{Q}_{1,2}^{(b)}(E_u) - \mathbb{Q}_{2,2}^{(a)}(E_g, 2) \mathbb{Q}_{1,1}^{(b)}(E_u) \right)}{2} \\
\text{z126} \quad \mathbb{Q}_3^{(c)}(A_u, 3) &= \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g, 2) \mathbb{Q}_{1,1}^{(b)}(E_u) + \mathbb{Q}_{2,2}^{(a)}(E_g, 2) \mathbb{Q}_{1,2}^{(b)}(E_u) \right)}{2} \\
\text{z127} \quad \mathbb{Q}_1^{(1,-1;c)}(A_u, a) &= \frac{\sqrt{10} \left(\sqrt{3} \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1) \mathbb{Q}_{1,2}^{(b)}(E_u) - \sqrt{3} \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1) \mathbb{Q}_{1,1}^{(b)}(E_u) + 2 \mathbb{Q}_2^{(1,-1;a)}(A_g) \mathbb{Q}_1^{(b)}(A_u) \right)}{10} \\
\text{z128} \quad \mathbb{Q}_1^{(1,-1;c)}(A_u, b) &= \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2} \\
\text{z129} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, 1a) &= \frac{\sqrt{5} \left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1) \mathbb{Q}_{1,2}^{(b)}(E_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1) \mathbb{Q}_{1,1}^{(b)}(E_u) + \sqrt{3} \mathbb{Q}_2^{(1,-1;a)}(A_g) \mathbb{Q}_1^{(b)}(A_u) \right)}{5} \\
\text{z130} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, 1b) &= \frac{\sqrt{2} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2} \\
\text{z131} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, 2a) &= \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2) \mathbb{Q}_{1,2}^{(b)}(E_u) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2) \mathbb{Q}_{1,1}^{(b)}(E_u) \right)}{2} \\
\text{z132} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, 2b) &= \frac{-\sqrt{2} \mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{2} \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{3} \mathbb{M}_3^{(1,-1;a)}(A_g, 3) \mathbb{T}_1^{(b)}(A_u)}{4} \\
\text{z133} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, 3a) &= \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2) \mathbb{Q}_{1,1}^{(b)}(E_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2) \mathbb{Q}_{1,2}^{(b)}(E_u) \right)}{2} \\
\text{z134} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, 3b) &= \frac{-\sqrt{2} \mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{2} \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{3} \mathbb{M}_3^{(1,-1;a)}(A_g, 2) \mathbb{T}_1^{(b)}(A_u)}{4} \\
\text{z135} \quad \mathbb{Q}_1^{(1,0;c)}(A_u, a) &= \frac{\sqrt{2} \left(\mathbb{G}_{1,1}^{(1,0;a)}(E_g) \mathbb{Q}_{1,2}^{(b)}(E_u) - \mathbb{G}_{1,2}^{(1,0;a)}(E_g) \mathbb{Q}_{1,1}^{(b)}(E_u) \right)}{2} \\
\text{z136} \quad \mathbb{Q}_1^{(1,0;c)}(A_u, b) &= \frac{\sqrt{10} \left(\sqrt{3} \mathbb{T}_{2,1}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3} \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + 2 \mathbb{T}_2^{(1,0;a)}(A_g) \mathbb{T}_1^{(b)}(A_u) \right)}{10} \\
\text{z137} \quad \mathbb{Q}_3^{(1,0;c)}(A_u, 1) &= \frac{\sqrt{5} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3} \mathbb{T}_2^{(1,0;a)}(A_g) \mathbb{T}_1^{(b)}(A_u) \right)}{5}
\end{aligned}$$

$$\boxed{\text{z138}} \quad \mathbb{Q}_3^{(1,0;c)}(A_u, 2) = \frac{\sqrt{2} \left(\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z139}} \quad \mathbb{Q}_3^{(1,0;c)}(A_u, 3) = \frac{\sqrt{2} \left(\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z140}} \quad \mathbb{Q}_1^{(1,1;c)}(A_u, a) = \mathbb{Q}_0^{(1,1;a)}(A_g) \mathbb{Q}_1^{(b)}(A_u)$$

$$\boxed{\text{z141}} \quad \mathbb{Q}_1^{(1,1;c)}(A_u, b) = \frac{\sqrt{2} \left(\mathbb{M}_{1,1}^{(1,1;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(1,1;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z142}} \quad \mathbb{G}_0^{(c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(a)}(E_g) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{1,2}^{(a)}(E_g) \mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_1^{(a)}(A_g) \mathbb{T}_1^{(b)}(A_u) \right)}{3}$$

$$\boxed{\text{z143}} \quad \mathbb{G}_2^{(c)}(A_u, a) = -\frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g, 1) \mathbb{Q}_{1,1}^{(b)}(E_u) + \mathbb{Q}_{2,2}^{(a)}(E_g, 1) \mathbb{Q}_{1,2}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z144}} \quad \mathbb{G}_2^{(c)}(A_u, b) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(a)}(E_g) \mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(a)}(E_g) \mathbb{T}_{1,2}^{(b)}(E_u) + 2\mathbb{M}_1^{(a)}(A_g) \mathbb{T}_1^{(b)}(A_u) \right)}{6}$$

$$\boxed{\text{z145}} \quad \mathbb{G}_0^{(1,-1;c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_1^{(1,-1;a)}(A_g) \mathbb{T}_1^{(b)}(A_u) \right)}{3}$$

$$\boxed{\text{z146}} \quad \mathbb{G}_2^{(1,-1;c)}(A_u, a) = -\frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1) \mathbb{Q}_{1,1}^{(b)}(E_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1) \mathbb{Q}_{1,2}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z147}} \quad \mathbb{G}_2^{(1,-1;c)}(A_u, b) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(E_u) + 2\mathbb{M}_1^{(1,-1;a)}(A_g) \mathbb{T}_1^{(b)}(A_u) \right)}{6}$$

$$\boxed{\text{z148}} \quad \mathbb{G}_2^{(1,-1;c)}(A_u, c) = \frac{\sqrt{7} \left(\sqrt{2} \mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{2} \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3} \mathbb{M}_3^{(1,-1;a)}(A_g, 1) \mathbb{T}_1^{(b)}(A_u) \right)}{7}$$

$$\boxed{\text{z149}} \quad \mathbb{G}_4^{(1,-1;c)}(A_u, 1) = \frac{\sqrt{7} \left(-\sqrt{6} \mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{6} \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + 4\mathbb{M}_3^{(1,-1;a)}(A_g, 1) \mathbb{T}_1^{(b)}(A_u) \right)}{14}$$

$$\boxed{\text{z150}} \quad \mathbb{G}_4^{(1,-1;c)}(A_u, 2) = \frac{-\sqrt{6} \mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{6} \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2) \mathbb{T}_{1,1}^{(b)}(E_u) + 2\mathbb{M}_3^{(1,-1;a)}(A_g, 3) \mathbb{T}_1^{(b)}(A_u)}{4}$$

$$\boxed{\text{z151}} \quad \mathbb{G}_4^{(1,-1;c)}(A_u, 3) = \frac{\sqrt{6}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{6}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\mathbb{M}_3^{(1,-1;a)}(A_g, 2)\mathbb{T}_1^{(b)}(A_u)}{4}$$

$$\boxed{\text{z152}} \quad \mathbb{G}_0^{(1,0;c)}(A_u) = \frac{\sqrt{3}\left(\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(E_u) + \mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(E_u) + \mathbb{G}_1^{(1,0;a)}(A_g)\mathbb{Q}_1^{(b)}(A_u)\right)}{3}$$

$$\boxed{\text{z153}} \quad \mathbb{G}_2^{(1,0;c)}(A_u, a) = \frac{\sqrt{6}\left(-\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(E_u) - \mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(E_u) + 2\mathbb{G}_1^{(1,0;a)}(A_g)\mathbb{Q}_1^{(b)}(A_u)\right)}{6}$$

$$\boxed{\text{z154}} \quad \mathbb{G}_2^{(1,0;c)}(A_u, b) = -\frac{\sqrt{2}\left(\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u)\right)}{2}$$

$$\boxed{\text{z155}} \quad \mathbb{G}_0^{(1,1;c)}(A_u) = \frac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_1^{(1,1;a)}(A_g)\mathbb{T}_1^{(b)}(A_u)\right)}{3}$$

$$\boxed{\text{z156}} \quad \mathbb{G}_2^{(1,1;c)}(A_u) = \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\mathbb{M}_1^{(1,1;a)}(A_g)\mathbb{T}_1^{(b)}(A_u)\right)}{6}$$

$$\boxed{\text{z245}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, 1a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g, 1)}{2}$$

$$\boxed{\text{z246}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, 1a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g, 1)}{2}$$

$$\boxed{\text{z247}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, 1b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g, 1)\mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z248}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, 1b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g, 1)\mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z249}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, 1c) = \frac{\sqrt{14}\left(-\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g, 2)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g, 2)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) + \mathbb{Q}_2^{(a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g, 1)\right)}{14}$$

$$\boxed{\text{z250}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, 1c) = \frac{\sqrt{14}\left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g, 2)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g, 2)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) + \mathbb{Q}_2^{(a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g, 1)\right)}{14}$$

$$\boxed{\text{z251}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, 1d) = \frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_g)\mathbb{M}_{1,2}^{(b)}(E_g)}{2}$$

$$\begin{aligned}
\boxed{\text{z252}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, 1d) &= -\frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_g)\mathbb{M}_{1,1}^{(b)}(E_g)}{2} \\
\boxed{\text{z253}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, 2a) &= \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z254}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, 2a) &= \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z255}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, 2b) &= \frac{-\mathbb{Q}_{2,1}^{(a)}(E_g, 1)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) + \mathbb{Q}_{2,2}^{(a)}(E_g, 1)\mathbb{Q}_{2,2}^{(b)}(E_g, 1)}{2} \\
\boxed{\text{z256}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, 2b) &= \frac{\mathbb{Q}_{2,1}^{(a)}(E_g, 1)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) + \mathbb{Q}_{2,2}^{(a)}(E_g, 1)\mathbb{Q}_{2,1}^{(b)}(E_g, 1)}{2} \\
\boxed{\text{z257}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, 2c) &= \frac{\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{M}_{1,1}^{(b)}(E_g) - \mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{M}_{1,2}^{(b)}(E_g)}{2} \\
\boxed{\text{z258}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, 2c) &= -\frac{\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{M}_{1,2}^{(b)}(E_g) + \mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{M}_{1,1}^{(b)}(E_g)}{2} \\
\boxed{\text{z259}} \quad \mathbb{Q}_{4,1}^{(c)}(E_g, 1) &= \frac{\sqrt{7} \left(\mathbb{Q}_{2,1}^{(a)}(E_g, 2)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) - \mathbb{Q}_{2,2}^{(a)}(E_g, 2)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) + 2\sqrt{3}\mathbb{Q}_2^{(a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) \right)}{14} \\
\boxed{\text{z260}} \quad \mathbb{Q}_{4,2}^{(c)}(E_g, 1) &= \frac{\sqrt{7} \left(-\mathbb{Q}_{2,1}^{(a)}(E_g, 2)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) - \mathbb{Q}_{2,2}^{(a)}(E_g, 2)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) + 2\sqrt{3}\mathbb{Q}_2^{(a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) \right)}{14} \\
\boxed{\text{z261}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z262}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, 1a) &= \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z263}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, 1b) &= \frac{\sqrt{14} \left(-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) + \mathbb{Q}_2^{(1,-1;a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) \right)}{14} \\
\boxed{\text{z264}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, 1b) &= \frac{\sqrt{14} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) + \mathbb{Q}_2^{(1,-1;a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) \right)}{14}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z265}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, 1c) &= \frac{\sqrt{2}\mathbb{M}_1^{(1,-1;a)}(A_g)\mathbb{M}_{1,2}^{(b)}(E_g)}{2} \\
\boxed{\text{z266}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, 1c) &= -\frac{\sqrt{2}\mathbb{M}_1^{(1,-1;a)}(A_g)\mathbb{M}_{1,1}^{(b)}(E_g)}{2} \\
\boxed{\text{z267}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, 1d) &= \frac{\sqrt{26}\left(\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{M}_{1,1}^{(b)}(E_g) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{M}_{1,2}^{(b)}(E_g) - \sqrt{3}\mathbb{M}_3^{(1,-1;a)}(A_g, 1)\mathbb{M}_{1,2}^{(b)}(E_g)\right)}{26} \\
\boxed{\text{z268}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, 1d) &= \frac{\sqrt{26}\left(-\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{M}_{1,2}^{(b)}(E_g) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{M}_{1,1}^{(b)}(E_g) + \sqrt{3}\mathbb{M}_3^{(1,-1;a)}(A_g, 1)\mathbb{M}_{1,1}^{(b)}(E_g)\right)}{26} \\
\boxed{\text{z269}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, 2a) &= \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z270}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, 2a) &= \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z271}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, 2b) &= \frac{-\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_{2,2}^{(b)}(E_g, 1)}{2} \\
\boxed{\text{z272}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, 2b) &= \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_{2,1}^{(b)}(E_g, 1)}{2} \\
\boxed{\text{z273}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, 2c) &= \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(E_g) - \mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(E_g)}{2} \\
\boxed{\text{z274}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, 2c) &= -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(E_g) + \mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(E_g)}{2} \\
\boxed{\text{z275}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, 2d) &= \frac{-\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{M}_{1,1}^{(b)}(E_g) + \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{M}_{1,2}^{(b)}(E_g) + \sqrt{15}\mathbb{M}_3^{(1,-1;a)}(A_g, 2)\mathbb{M}_{1,2}^{(b)}(E_g) + \sqrt{15}\mathbb{M}_3^{(1,-1;a)}(A_g, 3)\mathbb{M}_{1,1}^{(b)}(E_g)}{8} \\
\boxed{\text{z276}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, 2d) &= \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{M}_{1,2}^{(b)}(E_g) + \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{M}_{1,1}^{(b)}(E_g) - \sqrt{15}\mathbb{M}_3^{(1,-1;a)}(A_g, 2)\mathbb{M}_{1,1}^{(b)}(E_g) + \sqrt{15}\mathbb{M}_3^{(1,-1;a)}(A_g, 3)\mathbb{M}_{1,2}^{(b)}(E_g)}{8} \\
\boxed{\text{z277}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 1a) &= \frac{\sqrt{7}\left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) + 2\sqrt{3}\mathbb{Q}_2^{(1,-1;a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g, 1)\right)}{14}
\end{aligned}$$

$$\begin{aligned}
\text{z278} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 1a) &= \frac{\sqrt{7} \left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) + 2\sqrt{3}\mathbb{Q}_2^{(1,-1;a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) \right)}{14} \\
\text{z279} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 1b) &= \frac{\sqrt{13} \left(\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{M}_{1,1}^{(b)}(E_g) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{M}_{1,2}^{(b)}(E_g) + 2\sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_g, 1)\mathbb{M}_{1,2}^{(b)}(E_g) \right)}{26} \\
\text{z280} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 1b) &= -\frac{\sqrt{13} \left(\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{M}_{1,2}^{(b)}(E_g) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{M}_{1,1}^{(b)}(E_g) + 2\sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_g, 1)\mathbb{M}_{1,1}^{(b)}(E_g) \right)}{26} \\
\text{z281} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 2) &= \frac{-\mathbb{M}_3^{(1,-1;a)}(A_g, 2)\mathbb{M}_{1,2}^{(b)}(E_g) + \mathbb{M}_3^{(1,-1;a)}(A_g, 3)\mathbb{M}_{1,1}^{(b)}(E_g)}{2} \\
\text{z282} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 2) &= \frac{\mathbb{M}_3^{(1,-1;a)}(A_g, 2)\mathbb{M}_{1,1}^{(b)}(E_g) + \mathbb{M}_3^{(1,-1;a)}(A_g, 3)\mathbb{M}_{1,2}^{(b)}(E_g)}{2} \\
\text{z283} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, 3) &= \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{M}_{1,1}^{(b)}(E_g) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{M}_{1,2}^{(b)}(E_g) + \mathbb{M}_3^{(1,-1;a)}(A_g, 2)\mathbb{M}_{1,2}^{(b)}(E_g) + \mathbb{M}_3^{(1,-1;a)}(A_g, 3)\mathbb{M}_{1,1}^{(b)}(E_g)}{8} \\
\text{z284} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, 3) &= \frac{-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{M}_{1,2}^{(b)}(E_g) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{M}_{1,1}^{(b)}(E_g) - \mathbb{M}_3^{(1,-1;a)}(A_g, 2)\mathbb{M}_{1,1}^{(b)}(E_g) + \mathbb{M}_3^{(1,-1;a)}(A_g, 3)\mathbb{M}_{1,2}^{(b)}(E_g)}{8} \\
\text{z285} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, 1a) &= -\frac{\sqrt{2}\mathbb{G}_1^{(1,0;a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g, 1)}{2} \\
\text{z286} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, 1a) &= \frac{\sqrt{2}\mathbb{G}_1^{(1,0;a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g, 1)}{2} \\
\text{z287} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, 1b) &= \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_0^{(b)}(A_g)}{2} \\
\text{z288} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, 1b) &= \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_0^{(b)}(A_g)}{2} \\
\text{z289} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, 1c) &= \frac{\sqrt{10} \left(\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 2)\mathbb{M}_{1,1}^{(b)}(E_g) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 2)\mathbb{M}_{1,2}^{(b)}(E_g) + \sqrt{3}\mathbb{T}_2^{(1,0;a)}(A_g)\mathbb{M}_{1,1}^{(b)}(E_g) \right)}{10} \\
\text{z290} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, 1c) &= \frac{\sqrt{10} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 2)\mathbb{M}_{1,2}^{(b)}(E_g) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 2)\mathbb{M}_{1,1}^{(b)}(E_g) + \sqrt{3}\mathbb{T}_2^{(1,0;a)}(A_g)\mathbb{M}_{1,2}^{(b)}(E_g) \right)}{10}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z291}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, 2a) &= \frac{\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g, 1) - \mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g, 1)}{2} \\
\boxed{\text{z292}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, 2a) &= -\frac{\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g, 1) + \mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g, 1)}{2} \\
\boxed{\text{z293}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, 2b) &= \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z294}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, 2b) &= \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z295}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, 2c) &= \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 1)\mathbb{M}_{1,1}^{(b)}(E_g) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 1)\mathbb{M}_{1,2}^{(b)}(E_g)}{2} \\
\boxed{\text{z296}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, 2c) &= \frac{\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 1)\mathbb{M}_{1,2}^{(b)}(E_g) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 1)\mathbb{M}_{1,1}^{(b)}(E_g)}{2} \\
\boxed{\text{z297}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g, 1a) &= \frac{\sqrt{2}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g, 1)}{2} \\
\boxed{\text{z298}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g, 1a) &= \frac{\sqrt{2}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g, 1)}{2} \\
\boxed{\text{z299}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g, 1b) &= \frac{\sqrt{2}\mathbb{M}_1^{(1,1;a)}(A_g)\mathbb{M}_{1,2}^{(b)}(E_g)}{2} \\
\boxed{\text{z300}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g, 1b) &= -\frac{\sqrt{2}\mathbb{M}_1^{(1,1;a)}(A_g)\mathbb{M}_{1,1}^{(b)}(E_g)}{2} \\
\boxed{\text{z301}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g, 2) &= \frac{\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(E_g) - \mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(E_g)}{2} \\
\boxed{\text{z302}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g, 2) &= -\frac{\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(E_g) + \mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(E_g)}{2} \\
\boxed{\text{z303}} \quad \mathbb{G}_{1,1}^{(c)}(E_g) &= \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{T}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z304}} \quad \mathbb{G}_{1,2}^{(c)}(E_g) &= \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{T}_0^{(b)}(A_g)}{2}
\end{aligned}$$

$$\boxed{\text{z305}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(E_g) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z306}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(E_g) = \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z307}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z308}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(E_g, 1) = \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z309}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z310}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(E_g, 2) = \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z311}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(E_g, a) = \frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z312}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(E_g, a) = \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z313}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(E_g, b) = \frac{\sqrt{5} \left(\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 2)\mathbb{M}_{1,1}^{(b)}(E_g) - \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g, 2)\mathbb{M}_{1,2}^{(b)}(E_g) - 2\mathbb{T}_2^{(1,0;a)}(A_g)\mathbb{M}_{1,1}^{(b)}(E_g) \right)}{10}$$

$$\boxed{\text{z314}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(E_g, b) = -\frac{\sqrt{5} \left(\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 2)\mathbb{M}_{1,2}^{(b)}(E_g) + \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g, 2)\mathbb{M}_{1,1}^{(b)}(E_g) + 2\mathbb{T}_2^{(1,0;a)}(A_g)\mathbb{M}_{1,2}^{(b)}(E_g) \right)}{10}$$

$$\boxed{\text{z315}} \quad \mathbb{G}_{1,1}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{T}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z316}} \quad \mathbb{G}_{1,2}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{T}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z397}} \quad \mathbb{Q}_{1,1}^{(c)}(E_u, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{1,1}^{(b)}(E_u)}{2}$$

$$\begin{aligned}
\text{z398} \quad \mathbb{Q}_{1,2}^{(c)}(Eu, a) &= \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(Ag)\mathbb{Q}_{1,2}^{(b)}(Eu)}{2} \\
\text{z399} \quad \mathbb{Q}_{1,1}^{(c)}(Eu, b) &= \frac{\sqrt{5}\left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(Eg, 2)\mathbb{Q}_{1,1}^{(b)}(Eu) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(Eg, 1)\mathbb{Q}_1^{(b)}(Au) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(Eg, 2)\mathbb{Q}_{1,2}^{(b)}(Eu) - \mathbb{Q}_2^{(a)}(Ag)\mathbb{Q}_{1,1}^{(b)}(Eu)\right)}{10} \\
\text{z400} \quad \mathbb{Q}_{1,2}^{(c)}(Eu, b) &= \frac{\sqrt{5}\left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(Eg, 1)\mathbb{Q}_1^{(b)}(Au) - \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(Eg, 2)\mathbb{Q}_{1,2}^{(b)}(Eu) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(Eg, 2)\mathbb{Q}_{1,1}^{(b)}(Eu) - \mathbb{Q}_2^{(a)}(Ag)\mathbb{Q}_{1,2}^{(b)}(Eu)\right)}{10} \\
\text{z401} \quad \mathbb{Q}_{1,1}^{(c)}(Eu, c) &= \frac{\mathbb{M}_{1,2}^{(a)}(Eg)\mathbb{T}_1^{(b)}(Au) - \mathbb{M}_1^{(a)}(Ag)\mathbb{T}_{1,2}^{(b)}(Eu)}{2} \\
\text{z402} \quad \mathbb{Q}_{1,2}^{(c)}(Eu, c) &= \frac{-\mathbb{M}_{1,1}^{(a)}(Eg)\mathbb{T}_1^{(b)}(Au) + \mathbb{M}_1^{(a)}(Ag)\mathbb{T}_{1,1}^{(b)}(Eu)}{2} \\
\text{z403} \quad \mathbb{Q}_{3,1}^{(c)}(Eu, 1) &= \frac{\sqrt{5}\left(-\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(Eg, 2)\mathbb{Q}_{1,1}^{(b)}(Eu) - 4\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(Eg, 1)\mathbb{Q}_1^{(b)}(Au) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(Eg, 2)\mathbb{Q}_{1,2}^{(b)}(Eu) + 6\mathbb{Q}_2^{(a)}(Ag)\mathbb{Q}_{1,1}^{(b)}(Eu)\right)}{30} \\
\text{z404} \quad \mathbb{Q}_{3,2}^{(c)}(Eu, 1) &= \frac{\sqrt{5}\left(4\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(Eg, 1)\mathbb{Q}_1^{(b)}(Au) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(Eg, 2)\mathbb{Q}_{1,2}^{(b)}(Eu) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(Eg, 2)\mathbb{Q}_{1,1}^{(b)}(Eu) + 6\mathbb{Q}_2^{(a)}(Ag)\mathbb{Q}_{1,2}^{(b)}(Eu)\right)}{30} \\
\text{z405} \quad \mathbb{Q}_{3,1}^{(c)}(Eu, 2) &= \frac{\sqrt{6}\left(\mathbb{Q}_{2,1}^{(a)}(Eg, 1)\mathbb{Q}_{1,1}^{(b)}(Eu) - \mathbb{Q}_{2,2}^{(a)}(Eg, 1)\mathbb{Q}_{1,2}^{(b)}(Eu) - \mathbb{Q}_{2,2}^{(a)}(Eg, 2)\mathbb{Q}_1^{(b)}(Au)\right)}{6} \\
\text{z406} \quad \mathbb{Q}_{3,2}^{(c)}(Eu, 2) &= \frac{\sqrt{6}\left(-\mathbb{Q}_{2,1}^{(a)}(Eg, 1)\mathbb{Q}_{1,2}^{(b)}(Eu) + \mathbb{Q}_{2,1}^{(a)}(Eg, 2)\mathbb{Q}_1^{(b)}(Au) - \mathbb{Q}_{2,2}^{(a)}(Eg, 1)\mathbb{Q}_{1,1}^{(b)}(Eu)\right)}{6} \\
\text{z407} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(Eu, a) &= \frac{\sqrt{5}\left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(Eg, 2)\mathbb{Q}_{1,1}^{(b)}(Eu) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(Eg, 1)\mathbb{Q}_1^{(b)}(Au) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(Eg, 2)\mathbb{Q}_{1,2}^{(b)}(Eu) - \mathbb{Q}_2^{(1,-1;a)}(Ag)\mathbb{Q}_{1,1}^{(b)}(Eu)\right)}{10} \\
\text{z408} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(Eu, a) &= \frac{\sqrt{5}\left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(Eg, 1)\mathbb{Q}_1^{(b)}(Au) - \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(Eg, 2)\mathbb{Q}_{1,2}^{(b)}(Eu) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(Eg, 2)\mathbb{Q}_{1,1}^{(b)}(Eu) - \mathbb{Q}_2^{(1,-1;a)}(Ag)\mathbb{Q}_{1,2}^{(b)}(Eu)\right)}{10} \\
\text{z409} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(Eu, b) &= \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(Eg)\mathbb{T}_1^{(b)}(Au) - \mathbb{M}_1^{(1,-1;a)}(Ag)\mathbb{T}_{1,2}^{(b)}(Eu)}{2} \\
\text{z410} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(Eu, b) &= \frac{-\mathbb{M}_{1,1}^{(1,-1;a)}(Eg)\mathbb{T}_1^{(b)}(Au) + \mathbb{M}_1^{(1,-1;a)}(Ag)\mathbb{T}_{1,1}^{(b)}(Eu)}{2}
\end{aligned}$$

$$\boxed{\text{z411}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 1a) = \frac{\sqrt{5} \left(-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{1,1}^{(b)}(E_u) - 4\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_1^{(b)}(A_u) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{1,2}^{(b)}(E_u) + 6\mathbb{Q}_2^{(1,-1;a)}(A_g)\mathbb{Q}_{1,1}^{(b)}(E_u) \right)}{30}$$

$$\boxed{\text{z412}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 1a) = \frac{\sqrt{5} \left(4\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_1^{(b)}(A_u) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{1,1}^{(b)}(E_u) + 6\mathbb{Q}_2^{(1,-1;a)}(A_g)\mathbb{Q}_{1,2}^{(b)}(E_u) \right)}{30}$$

$$\boxed{\text{z413}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 1b) = \frac{-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{6}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_1^{(b)}(A_u) + \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - 6\mathbb{M}_3^{(1,-1;a)}(A_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u)}{12}$$

$$\boxed{\text{z414}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 1b) = \frac{-\sqrt{6}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_1^{(b)}(A_u) + \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 6\mathbb{M}_3^{(1,-1;a)}(A_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u)}{12}$$

$$\boxed{\text{z415}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 2a) = \frac{\sqrt{6} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_{1,1}^{(b)}(E_u) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_{1,2}^{(b)}(E_u) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_1^{(b)}(A_u) \right)}{6}$$

$$\boxed{\text{z416}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 2a) = \frac{\sqrt{6} \left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_{1,2}^{(b)}(E_u) + \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_1^{(b)}(A_u) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_{1,1}^{(b)}(E_u) \right)}{6}$$

$$\boxed{\text{z417}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(E_u, 2b) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{6}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_1^{(b)}(A_u) + 3\mathbb{M}_3^{(1,-1;a)}(A_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 3\mathbb{M}_3^{(1,-1;a)}(A_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u)}{12}$$

$$\boxed{\text{z418}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(E_u, 2b) = \frac{-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{6}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_1^{(b)}(A_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\mathbb{M}_3^{(1,-1;a)}(A_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\mathbb{M}_3^{(1,-1;a)}(A_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u)}{12}$$

$$\boxed{\text{z419}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(E_u, a) = \frac{\mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_1^{(b)}(A_u) - \mathbb{G}_1^{(1,0;a)}(A_g)\mathbb{Q}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z420}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(E_u, a) = \frac{-\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_1^{(b)}(A_u) + \mathbb{G}_1^{(1,0;a)}(A_g)\mathbb{Q}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z421}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(E_u, b) = \frac{\sqrt{5} \left(\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_1^{(b)}(A_u) - \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{T}_2^{(1,0;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{10}$$

$$\boxed{\text{z422}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(E_u, b) = \frac{\sqrt{5} \left(\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_1^{(b)}(A_u) - \sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{T}_2^{(1,0;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{10}$$

$$\boxed{\text{z423}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 1) = \frac{\sqrt{5} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 4\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_1^{(b)}(A_u) + \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 6\mathbb{T}_2^{(1,0;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{30}$$

$$\boxed{\text{z424}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 1) = \frac{\sqrt{5} \left(4\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_1^{(b)}(A_u) + \sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 6\mathbb{T}_2^{(1,0;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{30}$$

$$\boxed{\text{z425}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(E_u, 2) = \frac{\sqrt{6} \left(\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_1^{(b)}(A_u) \right)}{6}$$

$$\boxed{\text{z426}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(E_u, 2) = \frac{\sqrt{6} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{T}_{2,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_1^{(b)}(A_u) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{6}$$

$$\boxed{\text{z427}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(E_u, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z428}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(E_u, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z429}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(E_u, b) = \frac{\mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{T}_1^{(b)}(A_u) - \mathbb{M}_1^{(1,1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z430}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(E_u, b) = \frac{-\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{T}_1^{(b)}(A_u) + \mathbb{M}_1^{(1,1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z431}} \quad \mathbb{G}_{2,1}^{(c)}(E_u, 1a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g, 2)\mathbb{Q}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g, 1)\mathbb{Q}_1^{(b)}(A_u) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g, 2)\mathbb{Q}_{1,2}^{(b)}(E_u) + 3\mathbb{Q}_2^{(a)}(A_g)\mathbb{Q}_{1,1}^{(b)}(E_u)}{6}$$

$$\boxed{\text{z432}} \quad \mathbb{G}_{2,2}^{(c)}(E_u, 1a) = \frac{-\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g, 1)\mathbb{Q}_1^{(b)}(A_u) - \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g, 2)\mathbb{Q}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g, 2)\mathbb{Q}_{1,1}^{(b)}(E_u) + 3\mathbb{Q}_2^{(a)}(A_g)\mathbb{Q}_{1,2}^{(b)}(E_u)}{6}$$

$$\boxed{\text{z433}} \quad \mathbb{G}_{2,1}^{(c)}(E_u, 1b) = \frac{\mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{T}_1^{(b)}(A_u) + \mathbb{M}_1^{(a)}(A_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z434}} \quad \mathbb{G}_{2,2}^{(c)}(E_u, 1b) = -\frac{\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{T}_1^{(b)}(A_u) + \mathbb{M}_1^{(a)}(A_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z435}} \quad \mathbb{G}_{2,1}^{(c)}(E_u, 2a) = \frac{\sqrt{3} \left(-\mathbb{Q}_{2,1}^{(a)}(E_g, 1)\mathbb{Q}_{1,1}^{(b)}(E_u) + \mathbb{Q}_{2,2}^{(a)}(E_g, 1)\mathbb{Q}_{1,2}^{(b)}(E_u) - 2\mathbb{Q}_{2,2}^{(a)}(E_g, 2)\mathbb{Q}_1^{(b)}(A_u) \right)}{6}$$

$$\boxed{\text{z436}} \quad \mathbb{G}_{2,2}^{(c)}(E_u, 2a) = \frac{\sqrt{3} \left(\mathbb{Q}_{2,1}^{(a)}(E_g, 1)\mathbb{Q}_{1,2}^{(b)}(E_u) + 2\mathbb{Q}_{2,1}^{(a)}(E_g, 2)\mathbb{Q}_1^{(b)}(A_u) + \mathbb{Q}_{2,2}^{(a)}(E_g, 1)\mathbb{Q}_{1,1}^{(b)}(E_u) \right)}{6}$$

$$\begin{aligned}
\boxed{\text{z437}} \quad \mathbb{G}_{2,1}^{(c)}(Eu, 2b) &= \frac{\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{T}_{1,1}^{(b)}(Eu) - \mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{T}_{1,2}^{(b)}(Eu)}{2} \\
\boxed{\text{z438}} \quad \mathbb{G}_{2,2}^{(c)}(Eu, 2b) &= -\frac{\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{T}_{1,2}^{(b)}(Eu) + \mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{T}_{1,1}^{(b)}(Eu)}{2} \\
\boxed{\text{z439}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(Eu, 1a) &= \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{1,1}^{(b)}(Eu) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_1^{(b)}(Au) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{1,2}^{(b)}(Eu) + 3\mathbb{Q}_2^{(1,-1;a)}(A_g)\mathbb{Q}_{1,1}^{(b)}(Eu)}{6} \\
\boxed{\text{z440}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(Eu, 1a) &= \frac{-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_1^{(b)}(Au) - \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{1,2}^{(b)}(Eu) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_{1,1}^{(b)}(Eu) + 3\mathbb{Q}_2^{(1,-1;a)}(A_g)\mathbb{Q}_{1,2}^{(b)}(Eu)}{6} \\
\boxed{\text{z441}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(Eu, 1b) &= \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_1^{(b)}(Au) + \mathbb{M}_1^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(Eu)}{2} \\
\boxed{\text{z442}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(Eu, 1b) &= -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_1^{(b)}(Au) + \mathbb{M}_1^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(Eu)}{2} \\
\boxed{\text{z443}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(Eu, 1c) &= \frac{\sqrt{7}\left(\sqrt{30}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + 4\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_1^{(b)}(Au) - \sqrt{30}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - 3\sqrt{2}\mathbb{M}_3^{(1,-1;a)}(A_g, 1)\mathbb{T}_{1,2}^{(b)}(Eu)\right)}{42} \\
\boxed{\text{z444}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(Eu, 1c) &= \frac{\sqrt{7}\left(-4\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_1^{(b)}(Au) - \sqrt{30}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(Eu) - \sqrt{30}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(Eu) + 3\sqrt{2}\mathbb{M}_3^{(1,-1;a)}(A_g, 1)\mathbb{T}_{1,1}^{(b)}(Eu)\right)}{42} \\
\boxed{\text{z445}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(Eu, 2a) &= \frac{\sqrt{3}\left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_{1,1}^{(b)}(Eu) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_{1,2}^{(b)}(Eu) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_1^{(b)}(Au)\right)}{6} \\
\boxed{\text{z446}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(Eu, 2a) &= \frac{\sqrt{3}\left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_{1,2}^{(b)}(Eu) + 2\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2)\mathbb{Q}_1^{(b)}(Au) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1)\mathbb{Q}_{1,1}^{(b)}(Eu)\right)}{6} \\
\boxed{\text{z447}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(Eu, 2b) &= \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(Eu) - \mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(Eu)}{2} \\
\boxed{\text{z448}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(Eu, 2b) &= -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(Eu) + \mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(Eu)}{2} \\
\boxed{\text{z449}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(Eu, 2c) &= \frac{\sqrt{7}\left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(Eu) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(Eu) + \sqrt{30}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_1^{(b)}(Au) + 3\sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_g, 2)\mathbb{T}_{1,2}^{(b)}(Eu) + 3\sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_g, 3)\mathbb{T}_{1,1}^{(b)}(Eu)\right)}{42}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z450}} \quad & \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, 2c) \\
& = \frac{\sqrt{7} \left(\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{30}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_1^{(b)}(A_u) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - 3\sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{42} \\
\boxed{\text{z451}} \quad & \mathbb{G}_{4,1}^{(1,-1;c)}(E_u, 1) = \frac{\sqrt{7} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{30}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_1^{(b)}(A_u) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{28} \\
\boxed{\text{z452}} \quad & \mathbb{G}_{4,2}^{(1,-1;c)}(E_u, 1) = \frac{\sqrt{7} \left(-\sqrt{30}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_1^{(b)}(A_u) + \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - 2\sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{28} \\
\boxed{\text{z453}} \quad & \mathbb{G}_{4,1}^{(1,-1;c)}(E_u, 2) = \frac{-\mathbb{M}_3^{(1,-1;a)}(A_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_3^{(1,-1;a)}(A_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} \\
\boxed{\text{z454}} \quad & \mathbb{G}_{4,2}^{(1,-1;c)}(E_u, 2) = \frac{\mathbb{M}_3^{(1,-1;a)}(A_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_3^{(1,-1;a)}(A_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} \\
\boxed{\text{z455}} \quad & \mathbb{G}_{4,1}^{(1,-1;c)}(E_u, 3) = \frac{\sqrt{7} \left(\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) + 2\sqrt{6}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_1^{(b)}(A_u) - \mathbb{M}_3^{(1,-1;a)}(A_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \mathbb{M}_3^{(1,-1;a)}(A_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u) \right)}{28} \\
\boxed{\text{z456}} \quad & \mathbb{G}_{4,2}^{(1,-1;c)}(E_u, 3) = \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,2}^{(b)}(E_u) - 2\sqrt{6}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_1^{(b)}(A_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1)\mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{M}_3^{(1,-1;a)}(A_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_3^{(1,-1;a)}(A_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u) \right)}{28} \\
\boxed{\text{z457}} \quad & \mathbb{G}_{2,1}^{(1,0;c)}(E_u, 1a) = \frac{\mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_1^{(b)}(A_u) + \mathbb{G}_1^{(1,0;a)}(A_g)\mathbb{Q}_{1,2}^{(b)}(E_u)}{2} \\
\boxed{\text{z458}} \quad & \mathbb{G}_{2,2}^{(1,0;c)}(E_u, 1a) = -\frac{\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_1^{(b)}(A_u) + \mathbb{G}_1^{(1,0;a)}(A_g)\mathbb{Q}_{1,1}^{(b)}(E_u)}{2} \\
\boxed{\text{z459}} \quad & \mathbb{G}_{2,1}^{(1,0;c)}(E_u, 1b) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g, 1)\mathbb{T}_1^{(b)}(A_u) - \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) + 3\mathbb{T}_2^{(1,0;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{6} \\
\boxed{\text{z460}} \quad & \mathbb{G}_{2,2}^{(1,0;c)}(E_u, 1b) = \frac{-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 1)\mathbb{T}_1^{(b)}(A_u) - \sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u) - \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u) + 3\mathbb{T}_2^{(1,0;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{6} \\
\boxed{\text{z461}} \quad & \mathbb{G}_{2,1}^{(1,0;c)}(E_u, 2a) = \frac{\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(E_u) - \mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(E_u)}{2} \\
\boxed{\text{z462}} \quad & \mathbb{G}_{2,2}^{(1,0;c)}(E_u, 2a) = -\frac{\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(E_u) + \mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(E_u)}{2}
\end{aligned}$$

$$\boxed{\text{z463}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u, 2b) = \frac{\sqrt{3} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) - 2\mathbb{T}_{2,2}^{(1,0;a)}(E_g, 2) \mathbb{T}_1^{(b)}(A_u) \right)}{6}$$

$$\boxed{\text{z464}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u, 2b) = \frac{\sqrt{3} \left(\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,2}^{(b)}(E_u) + 2\mathbb{T}_{2,1}^{(1,0;a)}(E_g, 2) \mathbb{T}_1^{(b)}(A_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 1) \mathbb{T}_{1,1}^{(b)}(E_u) \right)}{6}$$

$$\boxed{\text{z465}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u, 1) = \frac{\mathbb{M}_{1,2}^{(1,1;a)}(E_g) \mathbb{T}_1^{(b)}(A_u) + \mathbb{M}_1^{(1,1;a)}(A_g) \mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z466}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u, 1) = -\frac{\mathbb{M}_{1,1}^{(1,1;a)}(E_g) \mathbb{T}_1^{(b)}(A_u) + \mathbb{M}_1^{(1,1;a)}(A_g) \mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z467}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u, 2) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(E_u) - \mathbb{M}_{1,2}^{(1,1;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z468}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u, 2) = -\frac{\mathbb{M}_{1,1}^{(1,1;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(E_u) + \mathbb{M}_{1,2}^{(1,1;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

Atomic SAMB

- bra: $\langle s, \uparrow |, \langle s, \downarrow |$
- ket: $|s, \uparrow\rangle, |s, \downarrow\rangle$

$$\boxed{\text{x1}} \quad \mathbb{Q}_0^{(a)}(A_g) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$\boxed{\text{x2}} \quad \mathbb{M}_1^{(1,-1;a)}(A_g) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & -\frac{\sqrt{2}}{2} \end{bmatrix}$$

$$\boxed{\text{x3}} \quad \mathbb{M}_{1,1}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$$

$$\boxed{\text{x4}} \quad \mathbb{M}_{1,2}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} \\ \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$$

- bra: $\langle s, \uparrow |, \langle s, \downarrow |$
- ket: $|p_x, \uparrow\rangle, |p_x, \downarrow\rangle, |p_y, \uparrow\rangle, |p_y, \downarrow\rangle, |p_z, \uparrow\rangle, |p_z, \downarrow\rangle$

$$\boxed{\text{x5}} \quad \mathbb{Q}_1^{(a)}(A_u) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$$

$$\boxed{\text{x6}} \quad \mathbb{Q}_{1,1}^{(a)}(E_u) = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x7}} \quad \mathbb{Q}_{1,2}^{(a)}(E_u) = \begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x8}} \quad \mathbb{Q}_1^{(1,0;a)}(A_u) = \begin{bmatrix} 0 & -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x9}} \quad \mathbb{Q}_{1,1}^{(1,0;a)}(E_u) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x10}} \quad \mathbb{Q}_{1,2}^{(1,0;a)}(E_u) = \begin{bmatrix} \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x11}} \quad \mathbb{G}_2^{(1,-1;a)}(A_u) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{6}}{12} & \frac{\sqrt{6}i}{6} & 0 \\ -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}i}{6} \end{bmatrix}$$

$$\boxed{\text{x12}} \quad \mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x13}} \quad \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x14}} \quad \mathbb{G}_{2,1}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x15}} \quad \mathbb{G}_{2,2}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} 0 & -\frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x16}} \quad \mathbb{G}_0^{(1,1;a)}(A_u) = \begin{bmatrix} 0 & \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \end{bmatrix}$$

$$\boxed{\text{x17}} \quad \mathbb{M}_2^{(1,-1;a)}(A_u) = \begin{bmatrix} 0 & -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{6} & 0 \\ -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$$

$$\boxed{\text{x18}} \quad \mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x19}} \quad \mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x20}} \quad \mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x21}} \quad \mathbb{M}_{2,2}^{(1,-1;a)}(E_u, 2) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x22}} \quad \mathbb{M}_0^{(1,1;a)}(A_u) = \begin{bmatrix} 0 & \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{6} & \frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{3}}{6} \end{bmatrix}$$

$$\boxed{\text{x23}} \quad \mathbb{T}_1^{(a)}(A_u) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \end{bmatrix}$$

$$\boxed{\text{x24}} \quad \mathbb{T}_{1,1}^{(a)}(E_u) = \begin{bmatrix} \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x25}} \quad \mathbb{T}_{1,2}^{(a)}(E_u) = \begin{bmatrix} 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x26}} \quad \mathbb{T}_1^{(1,0;a)}(A_u) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x27}} \quad \mathbb{T}_{1,1}^{(1,0;a)}(E_u) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & \frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x28}} \quad \mathbb{T}_{1,2}^{(1,0;a)}(E_u) = \begin{bmatrix} \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$$

- bra: $\langle p_x, \uparrow |, \langle p_x, \downarrow |, \langle p_y, \uparrow |, \langle p_y, \downarrow |, \langle p_z, \uparrow |, \langle p_z, \downarrow |$
- ket: $|p_x, \uparrow \rangle, |p_x, \downarrow \rangle, |p_y, \uparrow \rangle, |p_y, \downarrow \rangle, |p_z, \uparrow \rangle, |p_z, \downarrow \rangle$

$$\boxed{\text{x29}} \quad \mathbb{Q}_0^{(a)}(A_g) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \end{bmatrix}$$

$$\boxed{\text{x30}} \quad \mathbb{Q}_2^{(a)}(A_g) = \begin{bmatrix} -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} \end{bmatrix}$$

$$\boxed{\text{x31}} \quad \mathbb{Q}_{2,1}^{(a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x32}} \quad \mathbb{Q}_{2,2}^{(a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x33}} \quad \mathbb{Q}_{2,1}^{(a)}(E_g, 2) = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x34}} \quad \mathbb{Q}_{2,2}^{(a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \\ -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x35}} \quad \mathbb{Q}_2^{(1,-1;a)}(A_g) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & \frac{\sqrt{6}}{12} & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 \\ 0 & \frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x36}} \quad \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x37}} \quad \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x38}} \quad \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x39}} \quad \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x40}} \quad \mathbb{Q}_0^{(1,1;a)}(A_g) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x41}} \quad \mathbb{G}_1^{(1,0;a)}(A_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x42}} \quad \mathbb{G}_{1,1}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x43}} \quad \mathbb{G}_{1,2}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x44}} \quad \mathbb{M}_1^{(a)}(A_g) = \begin{bmatrix} 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x45}} \quad \mathbb{M}_{1,1}^{(a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x46}} \quad \mathbb{M}_{1,2}^{(a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x47}} \quad \mathbb{M}_1^{(1,-1;a)}(A_g) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$$

$$\boxed{\text{x48}} \quad \mathbb{M}_3^{(1,-1;a)}(A_g, 1) = \begin{bmatrix} -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & \frac{\sqrt{5}}{5} & 0 \\ -\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & -\frac{\sqrt{5}}{5} \end{bmatrix}$$

$$\boxed{\text{x49}} \quad \mathbb{M}_3^{(1,-1;a)}(A_g, 2) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x50}} \quad \mathbb{M}_3^{(1,-1;a)}(A_g, 3) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x51}} \quad \mathbb{M}_{1,1}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x52}} \quad \mathbb{M}_{1,2}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x53}} \quad \mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 1) = \begin{bmatrix} 0 & -\frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{60} & \frac{\sqrt{30}}{15} & 0 \\ -\frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & -\frac{\sqrt{30}}{15} \\ 0 & \frac{\sqrt{30}i}{60} & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 \\ -\frac{\sqrt{30}i}{60} & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 \\ \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{15} \\ 0 & -\frac{\sqrt{30}}{15} & 0 & 0 & \frac{\sqrt{30}}{15} & 0 \end{bmatrix}$$

$$\boxed{\text{x54}} \quad \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 1) = \begin{bmatrix} 0 & \frac{\sqrt{30}i}{60} & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 \\ -\frac{\sqrt{30}i}{60} & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{60} & 0 & \frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{15} & 0 \\ -\frac{\sqrt{30}}{60} & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{15} \\ 0 & 0 & \frac{\sqrt{30}}{15} & 0 & 0 & -\frac{\sqrt{30}i}{15} \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{15} & \frac{\sqrt{30}i}{15} & 0 \end{bmatrix}$$

$$\boxed{\text{x55}} \quad \mathbb{M}_{3,1}^{(1,-1;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x56}} \quad \mathbb{M}_{3,2}^{(1,-1;a)}(E_g, 2) = \begin{bmatrix} \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x57}} \quad \mathbb{M}_1^{(1,1;a)}(A_g) = \begin{bmatrix} -\frac{\sqrt{30}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} \\ 0 & \frac{\sqrt{30}}{30} & 0 & 0 & \frac{\sqrt{30}}{20} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & -\frac{\sqrt{30}i}{20} \\ 0 & 0 & 0 & \frac{\sqrt{30}}{30} & \frac{\sqrt{30}i}{20} & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{15} & 0 \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{15} \end{bmatrix}$$

$$\boxed{\text{x58}} \quad \mathbb{M}_{1,1}^{(1,1;a)}(E_g) = \begin{bmatrix} 0 & \frac{\sqrt{30}}{15} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{20} & 0 \\ \frac{\sqrt{30}}{15} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & -\frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 \\ \frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & 0 \\ \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{30} \\ 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{30}}{30} & 0 \end{bmatrix}$$

$$\boxed{\text{x59}} \quad \mathbb{M}_{1,2}^{(1,1;a)}(E_g) = \begin{bmatrix} 0 & \frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 \\ -\frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{15} & \frac{\sqrt{30}}{20} & 0 \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & \frac{\sqrt{30}i}{30} \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{30} & 0 \end{bmatrix}$$

$$\boxed{\text{x60}} \quad \mathbb{T}_2^{(1,0;a)}(A_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x61}} \quad \mathbb{T}_{2,1}^{(1,0;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \\ 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x62}} \quad \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 1) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{6} & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 \\ \frac{\sqrt{6}i}{6} & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & \frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & -\frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x63}} \quad \mathbb{T}_{2,1}^{(1,0;a)}(E_g, 2) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & \frac{\sqrt{6}i}{12} & 0 \\ -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ 0 & -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 \\ \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x64}} \quad \mathbb{T}_{2,2}^{(1,0;a)}(E_g, 2) = \begin{bmatrix} -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & -\frac{\sqrt{6}i}{12} & 0 \\ 0 & \frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 \\ \frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 \end{bmatrix}$$

Cluster SAMB

- Site cluster

** Wyckoff: 3e

$$\boxed{\text{y1}} \quad \mathbb{Q}_0^{(s)}(A_g) = \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3} \right]$$

$$\boxed{\text{y2}} \quad \mathbb{Q}_{2,1}^{(s)}(E_g, 1) = \left[\frac{\sqrt{14}}{14}, \frac{\sqrt{14}}{7}, -\frac{3\sqrt{14}}{14} \right]$$

$$\boxed{\text{y3}} \quad \mathbb{Q}_{2,2}^{(s)}(E_g, 1) = \left[-\frac{5\sqrt{42}}{42}, \frac{2\sqrt{42}}{21}, \frac{\sqrt{42}}{42} \right]$$

- Bond cluster

** Wyckoff: 6a@6g

$$\boxed{\text{y4}} \quad \mathbb{Q}_0^{(b)}(A_g) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6} \right]$$

$$\boxed{\text{y5}} \quad \mathbb{T}_0^{(b)}(A_g) = \left[\frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6} \right]$$

$$\boxed{\text{y6}} \quad \mathbb{Q}_1^{(b)}(A_u) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6} \right]$$

$$\boxed{\text{y7}} \quad \mathbb{T}_1^{(b)}(A_u) = \left[\frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6} \right]$$

$$\boxed{\text{y8}} \quad \mathbb{M}_{1,1}^{(b)}(E_g) = \left[\frac{\sqrt{7}i}{14}, \frac{\sqrt{7}i}{7}, -\frac{3\sqrt{7}i}{14}, \frac{\sqrt{7}i}{14}, \frac{\sqrt{7}i}{7}, -\frac{3\sqrt{7}i}{14} \right]$$

$$\boxed{\text{y9}} \quad \mathbb{M}_{1,2}^{(b)}(E_g) = \left[-\frac{5\sqrt{21}i}{42}, \frac{2\sqrt{21}i}{21}, \frac{\sqrt{21}i}{42}, -\frac{5\sqrt{21}i}{42}, \frac{2\sqrt{21}i}{21}, \frac{\sqrt{21}i}{42} \right]$$

$$\boxed{\text{y10}} \quad \mathbb{Q}_{2,1}^{(b)}(E_g, 1) = \left[\frac{\sqrt{7}}{14}, \frac{\sqrt{7}}{7}, -\frac{3\sqrt{7}}{14}, \frac{\sqrt{7}}{14}, \frac{\sqrt{7}}{7}, -\frac{3\sqrt{7}}{14} \right]$$

$$\boxed{\text{y11}} \quad \mathbb{Q}_{2,2}^{(b)}(E_g, 1) = \left[-\frac{5\sqrt{21}}{42}, \frac{2\sqrt{21}}{21}, \frac{\sqrt{21}}{42}, -\frac{5\sqrt{21}}{42}, \frac{2\sqrt{21}}{21}, \frac{\sqrt{21}}{42} \right]$$

$$\boxed{\text{y12}} \quad \mathbb{Q}_{1,1}^{(b)}(E_u) = \left[\frac{5\sqrt{21}}{42}, -\frac{2\sqrt{21}}{21}, -\frac{\sqrt{21}}{42}, -\frac{5\sqrt{21}}{42}, \frac{2\sqrt{21}}{21}, \frac{\sqrt{21}}{42} \right]$$

$$\boxed{\text{y13}} \quad \mathbb{Q}_{1,2}^{(b)}(E_u) = \left[\frac{\sqrt{7}}{14}, \frac{\sqrt{7}}{7}, -\frac{3\sqrt{7}}{14}, -\frac{\sqrt{7}}{14}, -\frac{\sqrt{7}}{7}, \frac{3\sqrt{7}}{14} \right]$$

$$\boxed{\text{y14}} \quad \mathbb{T}_{1,1}^{(b)}(E_u) = \left[\frac{5\sqrt{21}i}{42}, -\frac{2\sqrt{21}i}{21}, -\frac{\sqrt{21}i}{42}, -\frac{5\sqrt{21}i}{42}, \frac{2\sqrt{21}i}{21}, \frac{\sqrt{21}i}{42} \right]$$

$$\boxed{\text{y15}} \quad \mathbb{T}_{1,2}^{(b)}(E_u) = \left[\frac{\sqrt{7}i}{14}, \frac{\sqrt{7}i}{7}, -\frac{3\sqrt{7}i}{14}, -\frac{\sqrt{7}i}{14}, -\frac{\sqrt{7}i}{7}, \frac{3\sqrt{7}i}{14} \right]$$

Site and Bond

Table 5: Orbital of each site

#	site	orbital
1	A	$ s, \uparrow\rangle, s, \downarrow\rangle, p_x, \uparrow\rangle, p_x, \downarrow\rangle, p_y, \uparrow\rangle, p_y, \downarrow\rangle, p_z, \uparrow\rangle, p_z, \downarrow\rangle$

Table 6: Neighbor and bra-ket of each bond

#	head	tail	neighbor	head (bra)	tail (ket)
1	A	A	[1]	[s,p]	[s,p]

— Site in Unit Cell —

Sites in (conventional) cell (no plus set), SL = sublattice

Table 7: 'A' (#1) site cluster (3e), -1

SL	position ($\mathbf{s}$)	mapping
1	[0.50000, 0.00000, 0.00000]	[1,4]
2	[0.00000, 0.50000, 0.00000]	[2,5]
3	[0.50000, 0.50000, 0.00000]	[3,6]

— Bond in Unit Cell —

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 8: 1-th 'A'-'A' [1] (#1) bond cluster (6a@6g), D, $|\boldsymbol{v}|=0.5$ (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[-0.50000, -0.50000, 0.00000]	[0.25000, 0.75000, 0.00000]	[1]	(2,1)	(1,1)	[0,1,0]
2	[0.50000, 0.00000, 0.00000]	[0.25000, 0.50000, 0.00000]	[2]	(3,1)	(2,1)	[0,0,0]
3	[0.00000, 0.50000, 0.00000]	[0.50000, 0.75000, 0.00000]	[3]	(1,1)	(3,1)	[0,-1,0]
4	[0.50000, 0.50000, 0.00000]	[0.75000, 0.25000, 0.00000]	[4]	(2,1)	(1,1)	[-1,0,0]
5	[-0.50000, 0.00000, 0.00000]	[0.75000, 0.50000, 0.00000]	[5]	(3,1)	(2,1)	[1,0,0]
6	[0.00000, -0.50000, 0.00000]	[0.50000, 0.25000, 0.00000]	[6]	(1,1)	(3,1)	[0,0,0]